

Economic and Policy Analysis for Emission Reduction from the Brick Industry in Nepal

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Abstract

The brick industry is one of the primary sources of carbon dioxide emissions and local air pollutants in Nepal. Coal, which accounts for one-third of the current national carbon dioxide emissions from fossil fuel sources and is entirely imported, is the primary fuel in the brick industry. The brick industry accounts for 27 percent of the total carbon dioxide emissions from coal consumption. The adoption of clean technologies or fuels in the brick industry is crucial for improving air quality, enhancing energy independence, and meeting the country's nationally determined contribution under the Paris Climate Accord and the net-zero emission target set for 2045. Substitution of imported coal with domestic energy resources in the brick industry substantially reduces the country's import bills. This study examines the economics of various alternatives to reduce coal consumption and corresponding emissions from the brick industry. The study considers a range of carbon taxes

(US\$10 to US\$100 per ton of carbon dioxide), an environmental fiscal policy. The US\$10 per ton of carbon dioxide tax would increase brick production costs by 2 to 6 percent, depending on the energy efficiencies of the technologies. If the carbon tax were US\$100 per ton of carbon dioxide, the cost of bricks would increase by 12 to 36 percent. However, implementation of the policy may not be successful without enabling lower cost, clean alternatives. For example, replacing more coal with biomass provides direct cost and environmental savings but would require relaxing strict forest protections. The study recommends various promotional policies for non-fired alternative bricks. It also argues that since using electricity for firing bricks is an ideal option for reducing emissions from the brick industry in Nepal, the government and development partners should prioritize pilot projects for electric kilns.

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1. Introduction

Nepal depends on imports for its fossil fuel supplies. This implies that almost all fuel-based CO₂ emissions and most of the local air pollutants are caused by imported fuels. Coal, the most carbon-intensive fuel, currently accounts for 36% of the total national CO₂ emissions from fossil fuel combustion.³ Brick making is an energy-intensive process due to the high-temperature heat requirement and coal is the most common fuel used for heat in Nepal. The brick industry alone consumes more than one-third of the total coal supply in Nepal (ICIMOD, 2019a), and this industry accounts for 27% of the total national CO₂ emissions from coal consumption (Sadavarte et al., 2019). Several cities in Nepal are suffering from local air pollution as the pollution levels are many times higher than standards set by the World Health Organization (WHO). The capital city Kathmandu was the world's most polluted city with its Air Quality Index (AQI) 265 at 11:06 am on April 10, 2024. The fine particulate matter pollution (PM_{2.5}) level was 34 times the WHO's annual air quality guideline value at that time.⁴ Brick kilns are one of the sources of PM_{2.5} emissions in the city. The 2023 average PM_{2.5} concentration in Nepal was 8.5 times the WHO annual air quality guideline value.⁵

Nepal is experiencing rapid urbanization and post-earthquake reconstruction of buildings with rising demand for bricks. For example, between the last two census years (2011 and 2021), the number of housing units built with brick walls increased by 21% (CBS, 2014; NSO, 2023). If the current practice of brick manufacturing continues, CO₂ emissions from the brick industry will increase, thereby challenging Nepal's ability to meet its nationally determined contribution (NDC) by 2030⁶ and net-zero emission target by 2045. Moreover, local air pollution from the brick industry, particularly PM_{2.5}, causes major health problems. It is estimated that ambient PM_{2.5} caused 11,619 deaths in 2015, and around one-fifth of the total PM_{2.5} in the country is emitted from brick kilns (World Bank, 2019).

³ What are the main sources of CO₂ emissions in Nepal? <https://www.iea.org/countries/nepal/emissions#what-are-the-main-sources-of-co2-emissions-in-nepal>

⁴ "Kathmandu world's most polluted city, again". Kathmandu Post, April 10, 2024.

<https://kathmandupost.com/climate-environment/2024/04/10/kathmandu-world-s-most-polluted-city-again>.

⁵ <https://www.iqair.com/nepal>.

⁶ Nepal has not specified an emission reduction target under the NDC, instead it has specified activities to be undertaken. These include: (i) expanding clean power (mainly hydro) capacity from current 3,000 MW to 15,000 MW; (ii) 90% of all passenger vehicles sold to be electric; (iii) 25% of households to switch to electric cooking; and (iv) maintaining current 45% of the total area of the country covered by forest (MOFE, 2021).

In the absence of alternative fuels, coal use for firing clay bricks has been continuously increasing in the country. For example, between 2010 and 2022, coal imports increased by more than fourfold, from 244 million tons in 2009 to 1,248 million tons in 2022 (NTIP, 2023).⁷ Most coal is imported from India, Indonesia, the United States, South Africa, and Australia, and a very small quantity of low-grade lignite is domestically produced. Coal accounted for about 7% of the total fossil-fuel imports in Nepal in 2018 (CBS, 2022b).

The Government of Nepal has enacted several policies to reduce environmental and social damage from the brick industry. The key policies specific to the brick industry include i) the ban on traditional Bull Trench kilns with modern kilns, such as zig-zag, vertical shaft brick, tunnel, and Hoffman kilns, in 2009, and ii) the promotion of relatively energy-efficient vertical shaft brick kilns in 2010 (SMSEE, 2017). The government has also upgraded standards on emissions and stack height for different kiln types in 2018 from previously promulgated standards in 2008 (MOFE, 2018). In response to government regulations, brick kilns increasingly adopted a zig-zag technology system. The zig-zag technology reduced the emissions of suspended particulate matters (SPMs) from 700 mg/Nm³ (government-allowed rate) to 113 mg/Nm³.⁸ Likewise, the government's brick industry-specific directives/guidelines related to occupational safety and health (OSH) for workers in 2017 include i) not more than 8 hours of daily work with half an hour of rest after 5 hours of continuous work, ii) relatively shorter work duration for workers involved in the furnace and brick firing area, and iii) prioritization of dust control and the regulation of noise in the brick industry (ILO and MOLESS, 2022).

Despite the government's efforts to reduce the environmental and health externalities of the brick industry, the shift toward modern and climate-friendly brick production has been slow in Nepal. For example, the vertical shaft brick kiln (VSBK)⁹ represents less than 2% of all kilns and the remaining 1% of all kilns are Hoffman kiln (HK) and Tunnel Kiln (TK) (ICIMOD, 2019a). The limited response of the brick industry towards government regulations indicates that some

⁷ Imported coal is mainly used by the brick and cement industries, but the share of coal used by these industries in Nepal is not well documented because industries have the option to import coal directly from other countries (WECS, 2023).

⁸ "Brick kilns adopting zig-zag technology". The Himalayan. Dec 31, 2016. <https://thehimalayantimes.com/business/brick-kilns-adopting-zig-zag-technology>

⁹ Despite an effort from the government and international agencies (e.g., Swiss Agency for Development and Cooperation, Swiss Resource Center and Consultancies for Development, St. Gallen, Switzerland, and Development Alternatives, India) in transferring and helping to establish VSBK technology to clay-fired brick entrepreneurs in the early 2010s, its uptake has been low. A study by Eil et al. (2020) reported about 38 VSBKs in Nepal, including 3 in Kathmandu Valley, of which only 1 is in operation as of 2016. The study finds inferior quality of bricks produced and higher initial investment requirements compared to other brick kilns are the main reasons for the low uptake of the VSBKs.

other policy measures, such as pricing and fiscal measures, would be necessary to reduce CO₂ and local air pollutants from the brick industry. However, before the government considers pricing policies, it is necessary to understand the technical as well as economic feasibility of those policies. No rigorous and quantitative analyses are available for Nepal in this area. This study aims to contribute to filling the knowledge gaps.

The main objective of this study is to examine the economics of various technologies and fuels for clay-fired brick production, such as the adoption or retrofit of energy-efficient technologies, application of cleaner renewable (zero carbon) biomass fuels (e.g., sustainable fuel wood, wood pellets/chips, sawdust, and bagasse), and electric technology. This study also examines the economics of resource-efficient bricks¹⁰ (e.g., Compressed Stabilized Earth Blocks (CSEB) and Hollow Concrete Block (HCB)) and modification of brick-based buildings with other sustainable alternatives (e.g., wood-frame structures with plywood for roof, wall, and floor buildings) in reducing harmful air pollution by reducing demand for clay-fired bricks in the future. The study conducted economic analysis from both private and social perspectives based on a techno-economic analysis framework for the brick industry. The study also utilizes the information collected through a rapid field-visit survey, physical observations of 22 brick kilns selected across the country, and consultation with brick industry experts in Nepal (Timilsina and Malla, 2023).

The paper is organized as follows. Section 2 briefly introduces Nepal's brick industry, followed by methodological development in Section 3. Data collection through a survey is briefly discussed in Section 4. The main results along with sensitivity analyses are presented in Section 5. Section 6 presents policy implications/recommendations. Section 7 concludes the paper.

2. Overview of Nepal's Brick Industry

In 2018, there were about 1,349 operating brick kilns in Nepal,¹¹ and they produced an estimated 5.14 billion bricks, and the brick industry contributed 4% to the GDP, and employed nearly

¹⁰ Resource-efficient bricks here refer to bricks produced using sand, stone dust, cement, and soil mixed in different proportions and that do not go through the firing process. Since they are uncooked, their lifetime is shorter (20-30 years) as compared to the fired bricks which last hundreds of years.

¹¹ The brick sector in Nepal is a poorly regulated and unorganized sector, and informal in nature. The brick sector statistics, such as the number of registered brick kilns, their production volume and kiln technology types, the number of people involved, and the quantity of energy consumption and types, are in general not well documented. The task of capturing these data is challenging. For instance, there is a wide variation in number of brick kilns (registered or in operation) and volume of brick production statistics in Nepal. For example, ILO et al. (2020) reported 966 kilns producing 3.04 billion bricks in 2018, ICIMOD (2019a) reported 1,349 kilns producing 5.14 billion bricks in 2018, FNCCI (2017) estimated 1,100 kilns in operation in 2012 with a production

246,000 people (CBS, 2022a; ICIMOD, 2019a). A brief snapshot of Nepal’s brick industry is summarized in Table 1.

Table 1: Snapshot of Nepal’s brick industry

Parameter	Year	Value	Data Source
Number of brick kilns	2018	1,349	(ICIMOD, 2019a)
Distribution of kilns by province ^a	2018	P1 (8.8%), P2 (30.2%), P3 (21.2%), P4 (6.4%), P5 (23.4%) and P7 (9.8%)	(ILO et al., 2020)
Annual production	2018	5.14 billion	(ICIMOD, 2019a)
Value of annual output ^b	2020	NRs 14 billion	(CBS, 2022a)
Contribution to GDP	2020	4%	(CBS, 2022a)
Coal consumption	2018	504,750 tons	(ICIMOD, 2019a)
CO ₂ emissions	2016	2.2 million tons	(Eil et al., 2020)
Total industry employment ^c	2018	186,150	(ILO et al., 2020)
Brick price index (2015=100)	2021	127	(CBS, 2022b)
AAGR ^d of the construction industry	2013-22	6.3%	(CBS, 2022b)

Notes: ^a P1 is Koshi, P2 is Madhesh, P3 is Bagmati, P4 is Gandaki, P5 is Lumbini and P7 is Sudurpachim provinces. Karnali province reported only 2 kilns in the operation. ^b Based on the sum of three national classifications of industrial codes (NSIC), i.e., 2391, 2392, and 2393. ^c Out of total employment, 95% are manual workers and 5% are administrative workers. About half of the manual workers are of Indian origin. The total brick industry employment represents about 17% of the country’s total manufacturing industry workforce. ^d AAGR is the average annual growth rate.

The basic types of brick kilns for firing clay bricks in Nepal are the clamp kiln (CK), fixed chimney bull’s trench kiln (FC BTK), improved zigzag kiln (ZZK), vertical shaft brick kiln (VSBK), Hoffman kiln (HK), and tunnel kiln (TK). These kilns can be classified as *traditional* or *modern* kilns.¹² However, traditional kilns dominate all kilns, representing about 76% of all kilns (ICIMOD, 2019a). Among these types of brick kilns, the CK and FC BTK are the least energy efficient and most polluting while the TK is the most energy efficient and least polluting. It is possible that FC BTKs can be upgraded to improved ZZKs because of the similarity in their technical designs. However, very few FC BTK are converted to ZZK kilns due to a lack of skilled

capacity ranging from 15,000 to 50,000 bricks per day, Eil et al. (2020) reported 1,595 kilns producing 4.9 billion bricks in 2016, and DOI (2022) reported total number of registered kilns since 1993 as 1,577 at the end of 2018 and CBS (2022a) reported 1,008 registered brick kilns in 2018 but their information on how many of them are in operation is not available. Many factors contribute to these wide variations in the number of kilns, such as the operation of several non-registered brick kilns around the country, and poor government documentation.

¹² There are several classifications of clay-fired brick kilns, e.g., intermittent, and continuous kilns based on the production process, up-draught, down-draft, and cross-draft kilns based on airflow, and natural and induced or forced draft kilns based on the method of production Kumar and Maithel (2016). We classify brick kilns as either *traditional* or *modern* based on the combination of kiln technology differentiated by their specific energy consumption (SEC), i.e., the energy required in MJ to produce 1 kg of clay-fired brick, the quality of brick produced, brick quality, labor productivity, and emissions. In this study, six brick kilns are considered that are operated in Nepal. Two of these six kilns (CK and FC BTK) are considered as *traditional* kilns and the remaining four (ZZK, VSBK, HK and TK) are considered as *modern* kilns. Note that modern kilns have better SEC (fuel efficiency) and lower emissions, better brick quality and higher labor productivity than the traditional kilns. Similar terminologies are used by Eil et al. (2020). We consider ZZK, which is an improved and retrofitted version of FC BTK, as *modern*, because it can potentially achieve greater fuel efficiency due to optimized airflow and reduced heat loss (Maithel et al., 2013; Tibrewal et al., 2023).

workers for the construction and operation of ZZKs, and the limited access to financing the investment requirement for brick kiln owners.

The alternative bricks defined earlier have two types: *fly-ash* bricks and *solid* or *hollow* blocks) (Rawal et al., 2020). The fly-ash bricks are made in mechanized plants where a pan-mixer is used to mix fly-ash (a by-product of coal combustion) with sand, stone dust, lime, and cement to prepare the required blend of the mixture. Similar to fly-ash bricks, the solid or hollow blocks used in building construction are made in semi-mechanized or mechanized plants by mixing the required blend of raw materials, e.g., solid or hollow concrete block (mixture of cement, sand, and fine gravel), autoclaved aerated concrete (mixture of sand, gypsum, lime, cement, and aluminum powder), and compressed stabilized earth block (mixture of soil, sand, fine gravel and cement). The main difference between these alternative bricks is the composition of different types of raw materials. The main advantages of these alternative bricks are better chemical composition and a reduction in the environmental footprint (Vasić et al., 2021). These alternative bricks are gaining popularity in Nepal due to recent climate-related policies, however, their current actual production and share in the country's total bricks production are relatively small and limited to urban areas.

3. Methodology for the Economic Analysis

For this study, a techno-economic assessment that makes a distinction between the private and social costs of brick production is developed. The techno-economic assessment primarily focuses on the production phase, reflecting the perspective of a producer, and analyzes the technical and economic viability of switching to cleaner fuels from existing coal in brick production.

There are three primary system elements (preparation, mechanization, and baking) that are necessary to make the clay-fired brick. Process 1 (preparation) makes use of machine, electricity, diesel, animal, and human labor, process 2 (mechanization) makes use of machine, electricity, and human labor, and process 3 (baking or firing technology) makes use of fossil and biomass fuels. All three processes operate in a flow process over the year except for planned downtime (Figure 1). The costs associated with only activities that take place within the brick kiln areas are considered. Other activities and costs outside the kiln areas, such as transportation of raw materials

or final product costs, are excluded. For instance, in Figure 1, the cost estimation in this study is associated with only mechanization and baking technology.

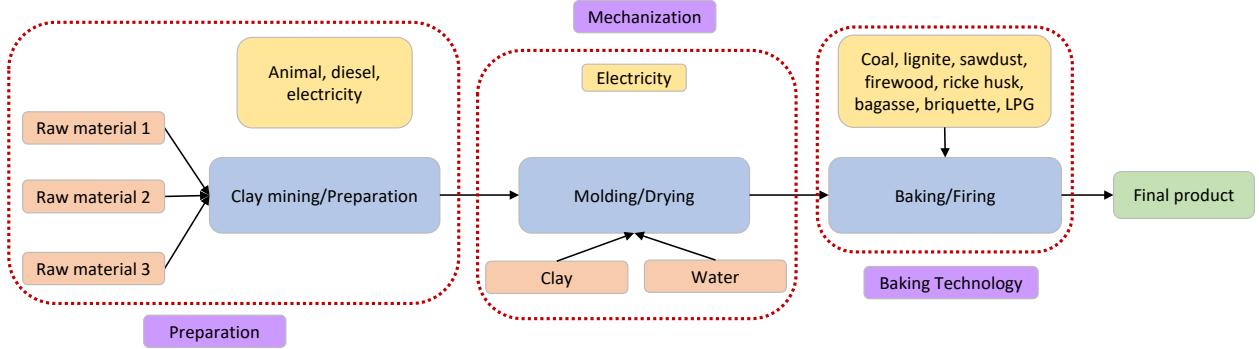


Figure 1: The brick-making processes

Economic analyses are conducted from multiple perspectives. First, the current production costs are compared by technologies and by provinces from a private (financial) perspective. This is followed by the comparison of costs from a social perspective. From the private perspective, the production costs include capital expenditures (e.g., costs of a kiln, land rental, and regulatory and compliance equipment costs), energy costs (e.g., coal, biomass, and electricity), labor costs, and the cost of materials (e.g., clay, water, and additives). The social production costs include environmental and health damage costs associated with brick production in addition to the private costs. The study also includes the economics of the substitution of coal with advanced biomass fuels for commonly used brick production technology (ZZK) and the production of alternative bricks. All these economic analyses use the information from the techno-economic assessment and the rapid survey conducted.

First, we estimated the private cost of producing clay-fired bricks in Nepal. The private cost of brick production ($C_{i,j}^P(t)$) for each province (i) and each technology type (j) in Nepal includes operational expenditures ($OpEx$) and capital expenditures ($CapEx$) for a particular year (t) (Equation 1).

$$C_{i,j}^p(t) = OpEx_{i,j}(t) + CapEx_{i,j}(t) \quad (1)$$

The operational expenditures [$OpEx_{i,j}(t)$] represent the ongoing manufacturing costs required to produce a brick (Equation 2). It includes direct and indirect costs. The direct costs

include the raw materials (e.g., clay, water, and energy costs) and labor costs, and the indirect costs include operation and maintenance costs, and utilities. The operational expenditure can also be divided into variable costs and fixed costs. Variable costs depend directly on the amount of product produced (e.g., raw materials) and fixed costs do not directly depend on the amount of product produced (e.g., labor). There are three main components of operational expenditures (*Equations 2.1-2.3*).

$$OpEx_{i,j}(t) = R_i(t) + L_i(t) + O\&M_{i,j}(t) \quad (2)$$

$$R_{i,j}(t) = \sum_k R_{i,j,k}(t) = R_{i,j,w}(t) + R_{i,j,c}(t) + \sum_e R_{j,e}(t) \quad (2.1)$$

where $k = \{w \text{ (water)}, c \text{ (clay)}, e \text{ (energy)}\}$

$$L_i(t) = \sum_i \frac{\text{average wage per brick } (t) \times \text{wage index}_i(t)}{\text{brick weight}_i(t)} + Others_i(t) \quad (2.2)$$

$$O\&M_{i,j}(t) = U_{i,j}(t) + M_{i,j}(t) \quad (2.3)$$

The first component is the raw materials cost $[R_{i,j}(t)]$ (*Equation 2.1*). The raw materials for brick production include water, clay, and different energy sources. The costs associated with water and clay raw materials that are used during the molding process and the costs associated with energy sources that are used during the baking (firing) process. The prices and the quantity of raw materials required for brick production are obtained from the survey and techno-economic assessment. The second component is the labor cost $[L_i(t)]$ which has two parts (*Equation 2.2*). The direct labor cost is estimated using a techno-economic assessment and the survey, such as the average wage and the weight of a brick, and a wage index that varies across the provinces (i). The indirect labor costs (others) such as purchasing of office supplies and advertisements, are excluded from the study. In this study, we assume the labor costs are independent of the technology used for brick production. The third component is the operation and maintenance costs $[O\&M_{i,j}(t)]$ (*Equation 2.3*). These costs include utilities $[U_{i,j}(t)]$, such as electricity for lighting and brick molding process but not for brick firing, and the maintenance cost of the brick kiln $[M_{i,j}(t)]$.

The capital expenditures $[CapEx_{i,j}(t)]$ include the fixed capital investment (e.g., land and chimney cost) and other cost items such as insurance and contingencies (*Equation 3*). We use a

simple financial function (PMT) to calculate the future capital expenditure payments for a loan, assuming constant payment and a constant cost of capital. The fixed capital investment for the land [$CapL_{i,j}(t)$] and chimney (technology) [$CapT_{i,j}(t)$] is estimated using the PMT function and the technical parameters (*Equations 3.1 and 3.2*). In PMT estimation, P_L and P_T are the periodic payment amounts for land and chimney, r is the cost of capital (interest rate), and n is the total number of payments over the lifetime of the brick-making plant. The technical parameters include the weight of the brick [$w_{i,j}(t)$] and the total number of brick productions in each year [$N_{i,j}(t)$]. However, in this study, the payment includes principal and interest but excludes taxes, reserve payments, or fees sometimes associated with the loans. The cost is calculated for the entire plant over 1 year when producing the specified annual production volume, leading to an NRs per year metric. The percentages of manufacturing overhead costs (insurance) are also calculated [$CapI_{i,j}(t)$] (*Equation 3.3*). It is estimated using the annual insurance percentage [$Irate_{i,j}(t)$] of land (L) and types of equipment (q) such as chimneys and other machines, and the technical parameters.

$$CapEx_{i,j}(t) = CapL_{i,j}(t) + CapT_{i,j}(t) + CapI_{i,j}(t) \quad (3)$$

$$CapL_{i,j}(t) = \frac{P_L \cdot r(1+r)^n}{(1+r)^n - 1} \div [(w_{i,j}(t) \cdot N_{i,j}(t))] \quad (3.1)$$

$$CapT_{i,j}(t) = \sum_q \frac{P_q \cdot r \cdot (1+r)^n}{(1+r)^n - 1} \div [(w_{i,j}(t) \cdot N_{i,j}(t))] \quad (3.2)$$

$$CapI_{i,j}(t) = \frac{Irate_{i,j}(t) \cdot \sum_q P_{L,q,i,j}(t)/n}{w_{i,j}(t) \cdot N_{i,j}(t)} \quad (3.3)$$

Second, as part of the sensitivity analysis, we also estimated the average cost of brick production [$C_j^p(t)$] under different coal prices and carbon taxes for commonly used brick-making technologies at the national level (*Equations 4 and 5*). The estimated average cost of brick production is simply the difference between the current cost of brick production [$\bar{C}_j^p(t)$] and the changes in cost due to coal prices [$\Delta C_j^p(t)$] and carbon taxes [$CT_{rate}(t) \cdot CE_j(t)$], where CT_{rate} is the carbon tax rate and the CE_j is the cost of carbon emissions.

$$C_j^p(\text{different coal price}) = \bar{C}_j^p(t) - \Delta C_j^p(t) \quad (4)$$

$$C_j^p(\text{different carbon tax}) = \bar{C}_j^p(t) + (CT_{rate}(t) \cdot CE_j(t)) \quad (5)$$

Third, we estimated changes in the energy cost of brick production $[C_i^p(t)]$ by partially substituting coal with advanced biomass fuels, such as pellets, for ZZK brick kilns across different provinces (*Equation 6*). The energy cost constitutes a major cost of brick production. The costs associated with a different combination of coal $[sh_{coal}]$ and pellets $[sh_{pellet}]$ is calculated using the fuel price information (fuel price index with the price of coal index as 1) collected from the rapid survey.

$$C_i^p(t) = C_i^p(coal) \cdot sh_{coal} + C_i^p(coal) \cdot fuel\ price\ index_i(t) \cdot sh_{pellet} \quad (6)$$

Fourth, we estimated the social cost of brick production. The social cost includes both private costs, i.e., direct costs to producers, and indirect or external costs to society that are not reflected in the market price. These external costs associated with brick production include environmental costs, such as costs associated with air pollution, GHG emissions, and soil degradation, health costs, such as workers' exposure to dust, and other societal impacts. In this study, the estimated social costs are limited to environmental costs $[C_{j,CO_2}^S(t)]$, which is the product of the environmental damage cost of CO₂ emissions from brick production using carbon tax as a proxy for the social cost of CO₂ $[SC_{CO_2}(t)]$ and the corresponding CO₂ emissions (*Equation 7*). The CO₂ emissions for brick production are estimated using the emission factor $[EF_{j,CO_2}(t)]$, specific energy consumption $[SEC_j(t)]$ and the average weight of the brick $[w_{i,j}(t)]$. Likewise, we estimated health damage costs associated with PM_{2.5} $[C_{i,j,PM2.5}^S(t)]$ using a similar approach as that for CO₂ emissions (*Equation 8*). The social cost values for CO₂ and PM_{2.5} are collected from existing literature.

$$C_{j,CO_2}^S(t) = EF_{j,CO_2}(t) \cdot SEC_j(t) \cdot w_{i,j}(t) \cdot SC_{CO_2}(t) \quad (7)$$

$$C_{i,j,PM2.5}^S(t) = EF_{i,j,PM2.5}(t) \cdot w_{i,j}(t) \cdot SC_{PM2.5}(t) \quad (8)$$

Lastly, we estimated the changes in costs of brick production using alternative non-fired bricks such as hollow concrete blocks (HCB) and compressed stabilized earth blocks (CSEB). We

compared the costs of fired and non-fired brick production [$C_m^B(t)$] based on the cost of building one square meter wall (Equation 9). These costs include material (brick) cost [$C_{brick}^B(t)$], labor cost [$C_{labor}^B(t)$], and raw material costs [$C_{raw\ material}^B(t)$]. The information needed for estimating these costs is collected from the available literature.

$$C_m^B(t) = C_{brick}^B(t) + C_{labor}^B(t) + C_{raw\ material}^B(t) \quad (9)$$

where $B = \{fired\ brick, non - fired\ bricks\ (CSEB\ and\ HCB)\}$

4. Data

Several different types of data exist that are relevant to this study, such as process-specific data, average data, and generic data. Each type of data is collected either from a primary or secondary source. For this study, the data and information are collected from two main sources. The first and primary source of data and information collected is from the field-visit rapid survey of 22 brick kilns selected across the country. The details of the survey, including the survey instruments and analysis, are available at (Timilsina and Malla, 2023).

To supplement data collected from primary sources through the rapid survey, several secondary sources were used. The data and information from the secondary sources are collected and compiled through comprehensive desk study and review of relevant peer-reviewed articles, and government and non-government published documents, reports, and articles. The key parameters considered in the techno-economic assessment and the cost estimation are summarized in Table 2. Selected information on key data collected from the survey is provided in Annex I and the detailed benchmark dataset description and sources are presented in Annex II.

Table 2: List of key parameters and data considered in the study

Study period	Base year: 2021
Kiln technology	CK, FC BTK, ZZK, VSBK, HK and TK
Fuel type	External fuel 1. Thermal energy for firing/baking: traditionally used fuels are coal, sawdust, fuelwood (chips), rick husk, bagasse, and biomass briquettes/pellets; 2. Electric energy for operating brick molding machine and induced (draft) fan.
Average brick weight	3.920 kg (Koshi); 3.375 kg (Madesh); 2.5 kg (Bagmati); 2.75 kg (Gandaki); 2.61 (Lumbini); 2.53 kg (Sudurpachim); and 2.88 kg (Nepal)
Conversion factor	1 kWh = 3.6 MJ; 1 toe = 41868 MJ; 1 kcal = 0.004184 MJ
Walling materials (Brick type)	Solid clay-fired bricks, Hollow concrete blocks (HCB) and Compressed stabilized earth blocks (CSEB)

Notes: CK refers to clamp kiln, FC BTK refers to fixed chimney bull's trench kiln, ZZK and VSBK are, respectively, improved zigzag kiln and vertical shaft brick kiln. HK and TK are Hoffman kiln and tunnel kiln, respectively.

5. Results from the Economic Analysis

5.1 Private costs of brick production

The brick production process is both energy and labor- intensive, and so are their costs. In addition to these operating costs, the capital expenditure costs (overhead costs) also contribute to the total cost of brick production. We discuss three different scenarios for analyzing the private costs associated with kilns: kilns that operate on coal only, a combination of coal and biomass fuels, and electricity.

5.1.1 Coal-based brick kilns

Most of the brick kilns in Nepal rely on imported low-quality coal, which is low in calorific value and high in ash content. The cost of imported coal accounts for a huge portion of the overall operating costs of brick production. equivalent to 30% to 40% of total brick production costs (SMSEE, 2017). The other most significant operating costs are labor costs. The manufacturing process also requires specialized chimneys and a large land area for raw materials, such as clay. The cost of chimneys and acquiring or leasing land adds significantly to the overhead costs of a brick factory. In addition to these three main cost components (coal, labor, and overhead), there are other less significant but important costs of brick production, including raw materials (water, sand), utilities, insurance premiums, maintenance expenses, transportation expenses, contingencies, and marketing and advertising expenses. Some of these additional costs, such as water as raw materials, utilities (electricity), insurance, and maintenance costs, are captured in overhead costs. The estimated total costs (energy, labor, and overhead costs) for producing clay-fired bricks by kiln technology and province are summarized in Table 3.

Table 3: Average total production cost per clay-fired brick by kiln technology and province (NRs/brick)

Kiln technology	Koshi	Madesh	Bagmati	Gandaki	Lumbini	Sudurpachim	Nepal
CK	30	27	28	24	25	24	27
FC BTK	18	17	20	15	17	16	18
ZZK	17	17	19	15	17	15	17

VSBK	14	14	17	13	15	13	15
HK	20	19	21	17	19	17	19
TK	45	44	46	42	43	42	44

Source: Field Survey (Timilsina and Malla, 2023)

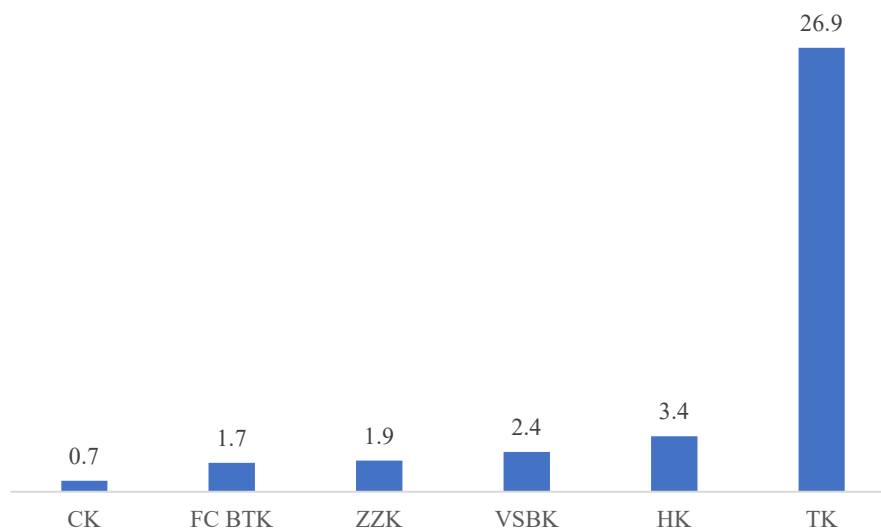


Figure 2. Average capital investment (chimney) cost by technology (NRs/brick)

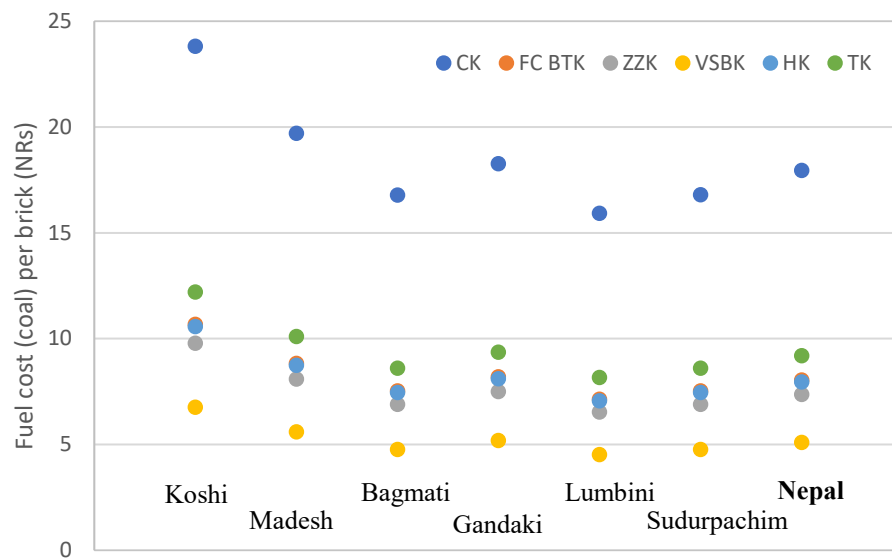


Figure 3. Average energy (coal) cost by kiln technology and province (NRs/brick)

At the technology level, the total cost of brick production varies widely across the provinces. The total brick production cost from TKs is the highest (NRs 44 per brick) and the VSBK is the lowest (NRs 15 per brick). Most of the operational brick kilns in the country are FC

BTK and ZZKs, and the total production costs from these kilns are about NRs 17 per brick, much lower compared to efficient TKs (Table 3). This is mainly due to the higher capital investment (chimney) cost of TK, about 14 times more than the investment cost of ZZK (Figure 2). Despite lower production costs and the technology promoted by the government in the 2010s, the use of VSBKs in the country is limited because the quality of bricks is poor, and the capital investment costs are higher compared to FC BTKs and ZZKs. Although the government has banned CKs, a small number of these kilns are in operation, and these kilns are the most energy-intensive. For example, the energy (coal) cost of producing a brick from CK is more than twice the cost of producing a brick from ZZK (Figure 3). The lowest energy cost of producing a brick is from VSBK, followed by ZZK, FC BTK, HK, TK, and CK, in increasing order across all the provinces (Figure 3). The VSBKs utilize waste heat from cooling bricks to preheat incoming raw bricks, which significantly improves energy efficiency. This heat recovery mechanism reduces the overall fuel consumption. Also, the vertical design of the kiln ensures a more uniform temperature distribution, leading to more efficient combustion and reduced fuel usage. The traditional kilns (e.g., CK and FC BTK) have poor heat transfer characteristics and require higher fuels than modern kilns (ZZK, VSBK, HK, and TK). For example, the specific energy consumption (SEC), a measure of energy efficiency of brick kiln technology measured in MJ of energy needed to produce 1 kg of brick, is much higher for traditional kilns when compared with modern kilns (Table A2.4). Higher energy requirements for traditional kilns contribute to higher energy costs. Almost three-fourths of all brick kilns are concentrated in three provinces (Madhesh, Lumbini, and Bagmati) (Figure 4). The brick sector statistics, such as kiln technology types are not well documented, however, half of all operational kilns are large in size by the number of people engaged, and FC BTK and ZZK are commonly used technologies by these large kilns.

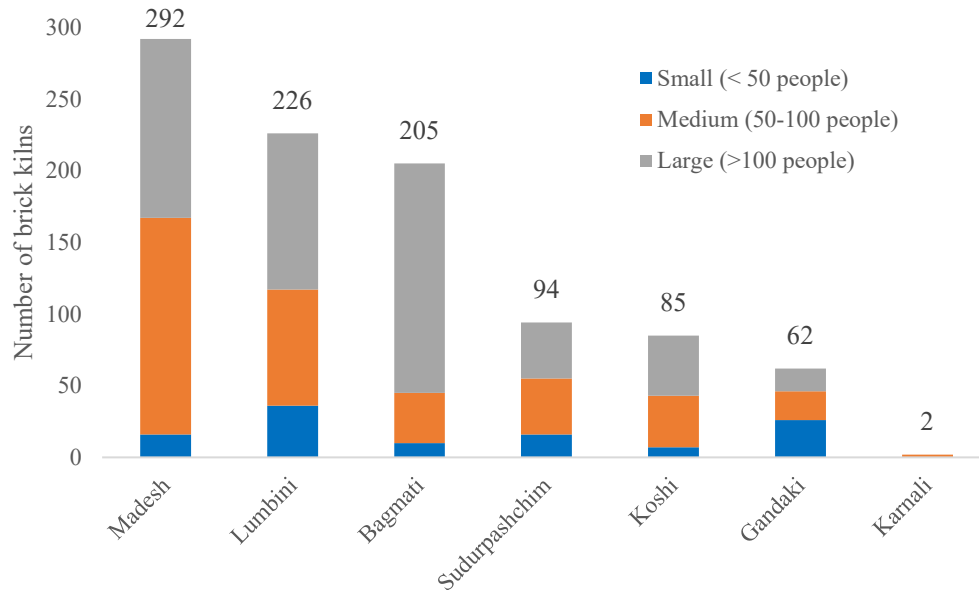


Figure 4. Number of brick kilns by size (people engaged) and province
Source: ILO et al. (2020)

At the province level, the estimated total brick production costs also vary widely, ranging from a high of NRs 46 per brick (TK) in Bagmati province to a low of NRs 15 per brick (VSBK) in Gandaki and Sudurpachim provinces (Table 3). However, the brick production cost ranges from commonly used FC BTK and ZZK across the provinces do not vary much, ranging from high NRs 20 per brick in Bagmati province to a low of NRs 15 per brick in Gandaki and Sudurpachim. Several factors contribute to variations in the cost of brick production across the provinces. For example, in Bagmati province, which includes Kathmandu Valley, coal prices, labor costs, and land rental costs are much higher than in other provinces, even though the average weight of the brick produced in the Bagmati province is relatively lower than in other provinces (Figure 5). Most of the brick kilns in Koshi, Madesh, Lumbini, and Sudurpachim provinces are in the plain terrains and close to the Indian border, and they have easy access to coal (imported from India) and Indian skilled labor (*Mistri*). These factors contribute to a relatively lower cost of brick production compared to brick kilns in Bagmati provinces with the same kiln technology. The average cost of energy (coal) ranges from 38% to 56% of the total costs, with a higher range in the eastern region and a lower range in the western region of the country. In Bagmati province, the average cost of energy is about one-third of the total costs. A similar result, the energy cost of 30% to 40% of total costs, is reported by (Sah et al., 2019) in Kathmandu Valley.



Figure 5. Average labor, land, and other production costs by province (NRs/brick)

The production costs are sensitive to values considered for input variables. Table 4 presents the results of sensitivity analyses on coal prices. Note that coal prices have almost doubled over the last three years, although the price might come down due to fluctuations in international prices. If the coal price is reduced, brick production cost is also expected to reduce assuming all other costs are the same. For example, if the coal price is reduced by 25% from the current price, the total production cost of brick is reduced from NRs 1 per brick for VSBK to NRs 4 for CK, equivalent reduction of costs of 10% (VSBK) to 17% (CK). As for the predominant kilns (FC BTK and ZZK) in the country, the total production cost is reduced by around NRs 2 per brick which is an equivalent of an 11% reduction of the total cost. Likewise, if the coal price is reduced further by 50% of the current price, the corresponding total brick production cost reduction is about NRs 4 per brick, an equivalent of a 21% reduction of the total cost. The reduction in the production cost due to lower coal prices is huge for kiln owners because the brick production process is energy intensive. The changes in total production cost are also estimated if the coal price is increased by 25% and 50% from the current price (Table 4). For FC BTK and ZZK, the total production cost increases by about NRs 2 and 4 per brick, an equivalent of an increase of 12% and 22% of total cost, respectively.

Table 4: Average brick production costs under different coal prices and carbon tax on coal (NRs/brick)

Kiln technology	Total cost (current)	Total cost due to change in coal price				Total cost due to carbon tax*			
		25% low	50% low	25% high	50% high	CT 10	CT 30	CT 50	CT100
CK	27	22	18	31	36	27.7	29.7	31.7	36.7
FC BTK	18	16	14	20	22	18.3	19.2	20.1	22.4
ZZK	17	16	14	19	21	17.8	18.6	19.4	21.5
VS BK	15	14	13	17	18	15.9	16.5	17.0	18.4
HK	19	17	15	21	23	19.9	20.7	21.6	23.9
TK	44	42	40	46	49	44.7	45.7	46.7	49.3

Notes: * Four carbon taxes (US\$ 10, 30, 50, and 100 per ton of CO₂ emissions) are considered. This translates into NRs 1.3, 3.9, 6.5, and 13 per kg of CO₂ emissions, assuming the current exchange rate of 1 US\$ = NRs 130.

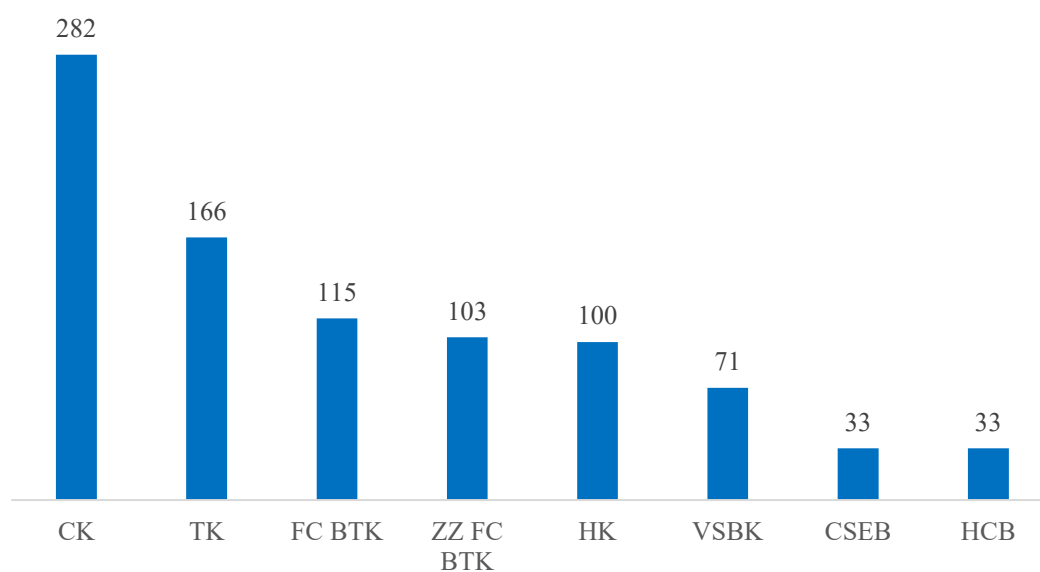


Figure 6. Average production-based emission intensity for coal-based brick kilns by technology type (g/kg of fired brick)

Although the government has set emission performance standards, they do not provide any incentive for brick industries to reduce emissions above and beyond the minimum that is required, even if the marginal costs of additional emission abatement are very small in comparison with the social costs they impose. However, economic instruments, such as the carbon tax on coal, incentivize brick industries to reduce CO₂ emissions and to innovate using cleaner fuels and processes. At the same time, imposing a carbon tax on coal used for brick production increases the total production cost. For example, if the carbon tax of US\$ 10 per ton of CO₂ is imposed on coal, the total cost of production increases from a high 4% for the least energy-efficient CK to a low 1%

for the most efficient TK (Table 4). Higher total costs of brick production are found with the higher carbon tax, e.g., with a carbon tax of US\$50 per ton of CO₂, the total cost of production for ZZK increases by 12%, and with a carbon tax of US\$100 per ton of CO₂, the production cost increases by 27%. A carbon tax on coal is one of many economic policy tools that can be implemented to encourage coal kiln owners to switch to cleaner fuels and modern kilns in the country. The carbon tax on coal-using brick kilns is also sensitive to the type of technologies used in brick-making. This is because the CO₂ emission intensity (e.g., gCO₂/kg of fired brick produced) varies widely by the types of technologies used for brick-making. For example, coal-based clamp kilns have the highest average production-based emission intensity, while VSBKs have the lowest CO₂ emission intensity (Figure 6). While TKs have advantages such as higher production capacity and more consistent firing results, they are more carbon-intensive than FC BTKs due to their higher fuel consumption and continuous operation. The non-firing blocks (CSEB and HCB) have the lowest CO₂ emission intensity.

5.1.2 Substitution of coal with biomass pellets for ZZK brick kilns

To reduce dependency on coal imports for brick production, various forms of alternate or renewable sources of energy must be developed. Biomass fuels, such as briquettes, pellets, wood chips, sawdust, and bagasse, in combination with coal, are commonly used in many traditional brick kilns in many South Asian countries. Biomass, such as pellets, is considered carbon-neutral because the CO₂ released during its combustion is offset by the CO₂ absorbed by the plants during their growth (assuming the plants are managed sustainably and not contributing to deforestation). When biomass is co-fired with coal, the overall carbon footprint would be smaller than that of burning coal alone. A survey study (Timilsina and Malla, 2023) observes that some of the brick kilns, particularly in the western part of Nepal, use biomass along with coal in the same kiln.¹³ However, large-scale use of biomass is constrained by its limited availability due to strict forest regulations (Timilsina and Malla, 2023). If the biomass supply constraint is relaxed and it is used along with coal at various proportions, the costs of brick production change. Table 5 presents the costs of brick production when part of the coal is substituted by carbon-neutral biomass fuels, such

¹³ Small amounts of biomass fuels are always used for the initial firing of the coal in most of the brick kilns (Timilsina and Malla, 2023).

as biomass pellets using ZZK technology. Rising coal prices in recent years are likely to further contribute to this shift in the coming years. In addition, a positive environmental impact works in favor of using biomass briquettes and pellets as a substitute for coal.

Table 5: Average energy costs per clay-fired brick for ZZK technology by province (NRs/brick)

	Koshi	Madhesh	Bagmati	Gandaki	Lumbini	Sudurpachim	Nepal
50% coal and 50% pellet	6.7	5.5	4.7	5.1	4.5	4.7	5.1
75% coal and 25% pellet	7.7	6.4	5.4	5.9	5.2	5.5	5.8

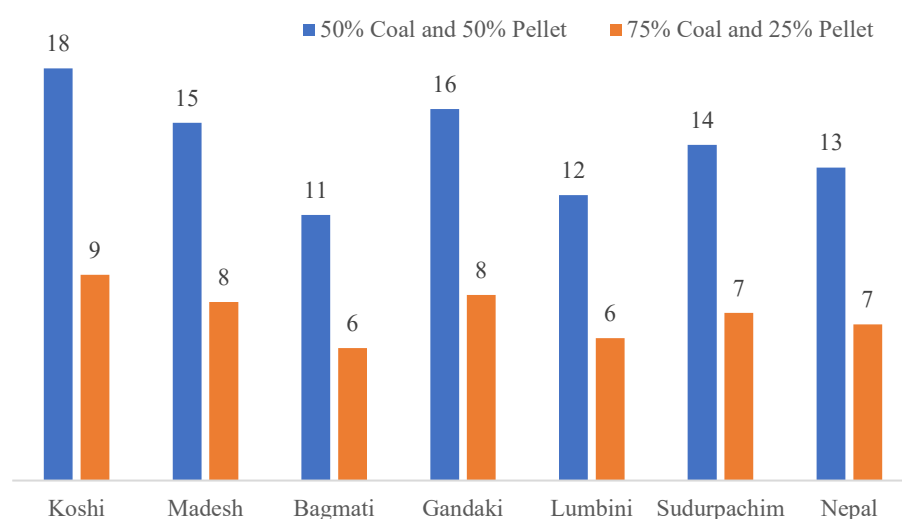


Figure 7. Percentage reduction in total costs of brick production using a combination of coal and biomass pellets compared to only coal by province (without carbon pricing)

Notes: These values are estimated based on the average price and the calorific value of coal as NRs 43.4 and 19.9 MJ per kg of coal, respectively, and the corresponding values for biomass pellets as NRs. 12.0 and 14.7 MJ per kg of pellet, respectively in 2022 based on information collected through a rapid survey by Timilsina and Malla (2023).

We estimated that the total cost of producing a brick can be reduced in the range of 6% to 9% across the provinces simply by substituting one-third of the coal with biomass pellets (Figure 7). The percentage reduction in brick production cost is much higher (11% to 18%) with a higher share of biomass pellets (50%). However, it may be unlikely that these levels of cost reduction can be achieved if the market for large-scale biomass fuels across the country does not exist. Clay-fired bricks can only be produced in a carbon-neutral manner if it is possible to completely replace the current primary energy source such as coal, with carbon-neutral energy sources, such as renewable biomass fuels. In our survey, we did find one brick kiln (FC BTK) in western Nepal

that uses 100% biomass briquettes imported from India but the kiln operator indicates importing it from India is challenging and it is not readily available in the domestic market.¹⁴ A study by (Lakho and Zardari, 2016) finds that the bricks fired with rapeseed husk and a mixture of sugarcane bagasse, rick husk, and used cloths at a commercial HK in Sind, Pakistan exhibited compressive strength of as high as 89% and 93% of those baked with coal, respectively. The saving in cost of bricks fired using these fuels ranges from 18% to 25% compared to those fired with coal. The rise in coal prices has been unprecedented in Nepal over the past few years mainly due to restrictions on low-priced coal export from India to its neighboring countries, high-priced coal import from overseas, and the Russian Federation's war with Ukraine. For example, between 2020 and 2022, the average CIF import price of coal per kg at the Indian border almost doubled from NRs 14 to 24 (Figure 8). In the past ten years or so, the share of coal imports from India also declined while it has increased from overseas (Figure 8). With additional government taxes and fees, and transportation costs, the market prices of coal reached as high as NRs 42 per kg by the end of 2021 (Prasain, 2021).

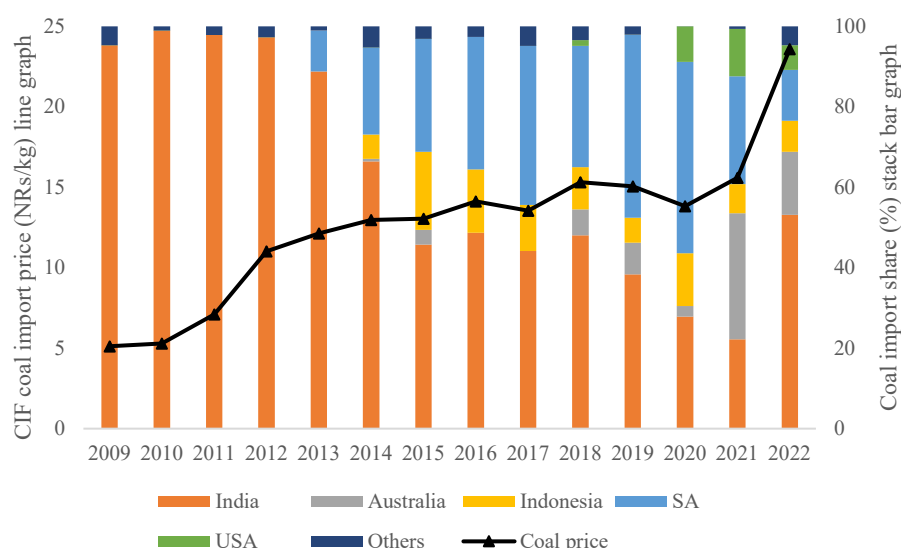


Figure 8. Import price of coal (line graph) and the share of coal imported from major countries to Nepal (stack graph).

Source: (NTIP, 2023)

¹⁴ In the field survey, technology-wise, most of the brick kilns are of natural and forced draft ZZK types (68%), followed by FC BTK (23%), and each one of CK and HK types (see Table A1.2). Coal is the dominant external fuel used for firing bricks. Out of the 22 surveyed kilns, 21 brick kilns use coal as the main fuel source for firing bricks. They also use a small amount of other fuels, mainly sawdust, rick husk, fuelwood chips, and bagasse, to supplement coal in firing bricks.

However, with clean energy and climate policies emerging, and renewable energy capacity in the electricity generation mix expanding, a recent study by IEA (IEA, 2023a) predicts that global demand for coal is expected to decline in the coming years. As demand declines, coal prices are also expected to fall in the future. To reflect these changes in coal prices, we did a simple sensitivity analysis of the reduction in total costs of brick production with different combinations of coal and biomass pellets (Figure 9). We find two key inferences from the sensitivity analysis. First, decreasing coal prices leads to a lower reduction in total costs and vice-versa with varying combinations of coal and pellets. Second, the higher the share of biomass pellets, the higher the reduction in total costs and vice-versa.

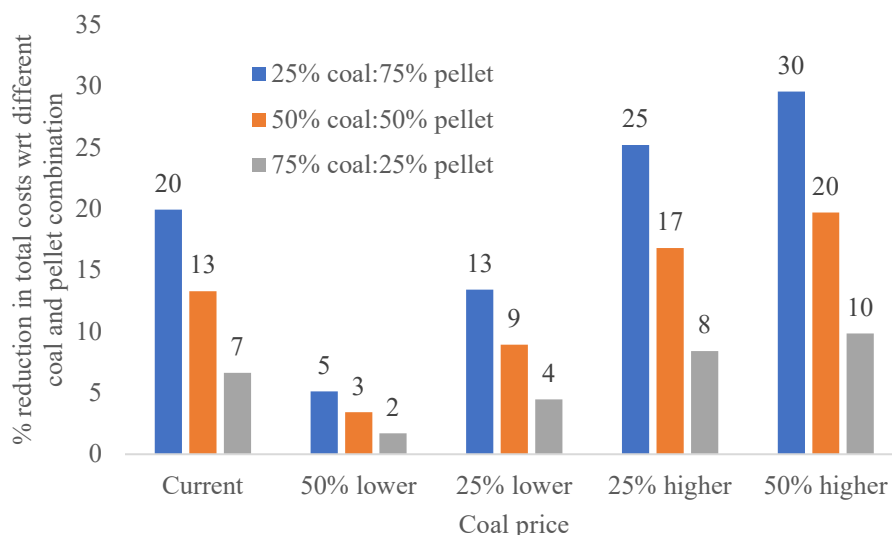


Figure 9. Variation in reduction in total brick production costs with different coal prices and the combination of coal and biomass pellets

5.1.3 Electric or hydrogen-based brick kilns

Since all fossil fuels are imported in Nepal, a promising environmentally friendly and carbon-neutral¹⁵ cost-effective alternative could be electric kilns powered by green electricity generated from renewable energy sources. Currently, electricity is commonly used for machine operation

¹⁵ Despite electric kilns powered by green electricity is carbon-neutral, there are significant process-related CO₂ emissions, due to the release of carbonates contained in the clay and the combustion of additives.

(molding), drying operation (dryer-oven), and utilities (lighting, HVAC) in the brick industry. Several advanced and fully automated TKs that use electricity for molding and drying are available in the market, but these kilns use either coal or natural gas for firing clay bricks.¹⁶ However, there are few recently built plants, such as the Kortemark production site in Belgium, which is considered the world's first climate-neutral ceramic plant operating on the highly efficient, electric kiln technology¹⁷ developed by Wienerberger company (Wienerberger, 2023). Further, as part of the *GreenBricks* project, Wienerberger company is also building a plant in Uttendorf (Austria) with an electric furnace, by eliminating the gas-fired burner with a modern, electrically powered high-temperature furnace, for brick production (NEFI, 2023). The plant is expected to be fully operational from 2025 onwards. With 100% green electricity being used, the plant is expected to reduce CO₂ emissions by approximately 90% and total energy requirements by 30% from the production of bricks. It is also reported that a roofing tile, a backing brick, or a clay paver kiln has an average power consumption of 0.4 kWh/kg brick product and is heated electrically.¹⁸

However, these electric kiln technologies are only at the demonstration stage. The electric kiln brick production process is fundamentally different. Clay-fired bricks can only be produced in a carbon-neutral manner if it is possible to completely replace the current primary energy sources for firing, such as coal or natural gas, with carbon-neutral energy sources, such as renewable-based biomass fuels or hydrogen. A study by (Geres et al., 2021) presented a viable path for GHG-neutral brick production in Germany with associated measures and costs by 2050. One of the proposed paths in their study is the possibility of using carbon-neutral hydrogen furnaces or electric kilns to produce clay-fired bricks in the future. However, the implementation of such electric kilns depends on massive energy and investment costs, about 2.35 billion euros over 30 years. At the present stage, such a huge cost associated with an electric kiln is less likely to be practical for developing countries like Nepal. However, policy changes, such as a carbon tax on coal and the promotion of green hydrogen from Nepal's growing hydropower production, and

¹⁶ See e.g., <https://cnbrickmachine.en.made-in-china.com/product/JobQTfxPTXRW/China-New-Design-Modern-Tunnel-Kiln-for-Clay-Brick-Production-Line.html>. This TK uses electricity for drying and coal or natural gas for firing (oven). The reference free-on-board price listed for this kiln ranges from US\$500,000 to US\$800,000.

¹⁷ In this plant, both the dryer and the kiln used to produce climate-neutral bricks are running entirely on green electricity, i.e., energy from fossil sources is not used at all.

¹⁸ https://www.zi-online.info/en/artikel/zi_The_electric_tunnel_kiln_challenge_heat_transfer-3464349.html

increasing fossil fuels import bill in the future will likely reduce the cost of producing hydrogen energy.

5.2 Social costs of clay-fired brick production

In addition to private costs, social costs include costs associated with health damage from hazardous air pollutants, e.g., PM_{2.5}, environmental damage from CO₂, material damage to buildings and structures, degradation of forest, soil, and water, and damage to crops. This study focuses on the health damage costs associated with PM_{2.5} and the environmental damage cost of CO₂ (carbon tax). The share of other damage costs in the total social costs is quite small and they are excluded in this study.

Burning low-grade coal and biomass fuels in traditional brick kilns is the major source of hazardous air pollutants, such as PM, BC, SO₂, NO_x, CO, VOCs, CO₂, and metals. Clay preparation (excavation and grinding) and firing are the two main stages that emit these emissions from brick kilns. Complete combustion of coal is possible only in the presence of an adequate supply of oxygen. However, not all coal is combusted 100% in the brick kilns. A large part of PM and CO emissions from brick kilns comes from incomplete combustion of coal and biomass in the kiln, while CO₂ emissions come from complete combustion. In controlled clinical trials, several studies find that higher levels of pulmonary dysfunctions, increased oxidative stress, and increased DNA damage in brick kiln workers than in non-brick kiln workers in India (Budhwar et al., 2003; Kaushik et al., 2012); (Tandon et al., 2017), and in Pakistan (Khisroon et al., 2018); (Raza and Ali, 2021; Raza et al., 2014). A study by (Shriraam et al., 2020) finds that one in ten migrant brick kiln workers in South India has symptoms suggestive of TB. A study by (Siraz et al., 2023) finds that coal-fired brick kiln workers and nearby residents on the outskirts of Dhaka City in Bangladesh are exposed to terrestrial radiation, although the levels of radiation are well below the recommended limit. In Nepal, brick workers report a higher prevalence of respiratory symptoms and illnesses related to exposure to ambient PM, such as chronic obstructive pulmonary disease (COPD), than non-brick workers (Sanjel et al., 2017).

Table 6: Number of deaths and cost of health damages from PM_{2.5} exposure in Nepal in 2019

	2019	2015*
Average annual exposure to ambient PM2.5	83.1 $\mu\text{g}/\text{m}^3$	
No. of deaths from PM2.5 exposure per 100,000 population	130	
Deaths from PM2.5 exposure as a share of all deaths	20.0%	
Annual cost of health damages from PM2.5 (% of GDP)	10.2%	
Death from PM2.5 associated with ambient air pollution (AAP)	17,948 (21,603)	11,619
Annual cost of health damages from AAP PM2.5 exposure	\$ 4,645 (5,723) million, PPP	\$ 686 million

Notes: The figure in the parenthesis is for indoor household air pollution (HAP). Source: (IHME, 2020) * The World Bank (2019) estimated the social cost of PM2.5 per mortality at US\$20,900–\$74,200 for 2015 in PPP.

It is estimated that brick kilns are responsible for more than one-fourth of total PM10 concentrations and about 40% of total black carbon emissions in the winter season in Kathmandu Valley, the time most of the brick kilns are in operation (Eil et al., 2020). Most deaths related to air pollution are caused by human exposure to PM2.5 (World Bank, 2021).¹⁹ Based on data from the Institute for Health Metrics and Evaluation's (IHME) Global Burden of Disease (GBD), air pollution ranks as the second highest risk factor, after malnutrition, which drives the most deaths in Nepal. The IHME GDB study also estimated the number of deaths and cost of health damages from PM2.5 exposure in Nepal (Table 6). The number of deaths from PM2.5 exposure in Nepal is highest in South Asia. Further, the average annual ambient (outdoor) concentrations of PM2.5 reached 83 $\mu\text{g}/\text{m}^3$ in 2019, which is way above the WHO guidelines of 10 $\mu\text{g}/\text{m}^3$. Most of the deaths associated with ambient PM2.5 are from heart disease, stroke, COPD, lung cancer, and ALRI (World Bank, 2021).

Assuming one-fourth of the country's total ambient air pollution (AAP) related PM2.5 emissions are attributed to brick production, the annual cost of health damages from PM2.5 exposure from brick kilns is estimated at US\$ 1,161 million in PPP and the annual number of deaths at 4,487 in 2019 (Table 6). This makes PM2.5 exposure one of the main risk factors for the number of deaths after malnutrition and before high blood pressure, dietary risks, tobacco smoking, and diabetes, among dozens of risk factors assessed by the GBD 2019 study.

The estimated social cost of PM2.5 varies by kiln technology and by province. In general, the social cost of PM2.5 from brick production is higher for less energy-efficient traditional kilns and lower for more energy-efficient modern kilns (Table 7). For example, the average social cost of brick production from FC BTK is almost double when compared to ZZK. Further, the social

¹⁹ Fine inhalable PM (PM2.5) comprises a portion of PM10. A recent ambient air quality monitoring study (Islam et al., 2022) shows PM2.5 to PM10 ratios ranges from 0.50 to 0.57 in Kathmandu Valley.

cost of PM2.5 is higher in the eastern regions than in the western regions of the country. This is due to the larger size of brick production in the eastern region than in the western region. The social cost of PM2.5 is associated with the health damage cost. It is important to note that an improvement in AAP (PM2.5) is, however, unlikely to provide immediate benefits for health outcomes (e.g., for heart disease, stroke, COPD, and lung cancer) that develop over long periods of PM2.5 exposure.

Table 7: Social costs of PM2.5 and CO₂ emissions of coal-based brick production in Nepal (NRs per brick)

Kiln technology	PM2.5							CO2
	Koshi	Madesh	Bagmati	Gandaki	Lumbini	Sudurpachim	Nepal	Nepal
CK	43	37	27	30	29	28	31	4
FC BTK	9	8	6	7	6	6	7	2
ZZK	5	5	3	4	4	3	4	2
VSBK	5	4	3	3	3	3	4	1
HK	7	6	4	5	4	4	5	2
TK	8	7	5	6	5	5	6	2

Notes: The estimated figures are the product of the emission factor, the average weight of the brick, and social cost. For PM2.5, e.g., it is the product of the emission factor (g/kg of brick), the average weight of the brick (kg/brick), and the social cost (US\$/kg). The social cost of PM2.5 is US\$ 86.85/kg and CO₂ is US\$ 0.04/kg taken from EPIC India, 2018).

Likewise, the social cost of CO₂ varies by kiln technology, with higher costs associated with less efficient and more energy-intensive technologies (e.g., CK) and lower costs associated with more efficient and less energy-intensive technologies (e.g., VSBK). The social cost of CO₂ is associated with the environmental damage cost.

5.3 Economics of alternative bricks

The devastating earthquake of 2015 and its aftershocks caused widespread damage to both life and property across the country, with an estimated 602,000 houses fully damaged, 285,000 houses partially damaged, and the loss of life was about 9,000 (DUDBC, 2017). As part of the government's *Build Back Better Initiative*, the government proposed several cost-efficient alternative materials for sustainable reconstruction of both urban and rural houses, such as interlocking brick, confined hollow concrete block, hollow concrete block (HCB), compressed

stabilized earth block (CSEB), random rubble with GI wire containment, and bamboo and stone hybrid Structure (DUDBC, 2017). Non-fired bricks are an environmentally friendly option as they do not require the burning of coal or other fossil fuels. Among these, two non-fired bricks (CSEB and HCB) have the most potential for growth in the country (Fordham, 2021).

Table 8: Cost of building one square meter wall with non-fired alternative bricks and clay-fired brick and cost savings in 2018 (NRs)

	Cost for 1 m ² wall (NRs)			Cost savings (%)	
	CSEB	HCB	Brick	CSEB	HCB
Material (bricks)	1,600	640	2,100	24	70
Labor	188	188	270	31	31
Raw materials	276	276	890	69	69
Total	2,064	1,104	3,260	37	66

Notes: CSEB is from Build Better Nepal (2018). For HCB, the material cost is from (MinErgy, 2020) and the other costs are the same as those of CSEB.

To assess the cost of different building materials, we calculated the costs of building a 1 square meter wall, using alternative bricks (CSEB and HCB) and clay-fired bricks. A striking contrast between alternative bricks and conventional clay-fired brick is the overall cost of building construction (Table 8). We find that the associated total building cost savings are about 37% with CSEB and much higher about 66% with HCB when compared with clay-fired bricks. It is noteworthy that most of the cost savings in absolute value come from the associated costs with material (bricks). This is important because the costs of building materials are the major cost component of the overall building construction of houses in the country. Despite these cost-saving benefits, the current use of CSEB and HCB relative to clay-fired brick for building materials for houses remains limited. The combination of better aesthetic value and longer life of red clay-fired bricks over CSEB and HCB is limiting the use of these alternative brick materials. Other important factors for the low adoption of alternative bricks in the country include a lack of marketing initiatives and national standards, inadequate consumer knowledge and competent labor, the attitude of stakeholders toward alternative bricks, and low-profit margins (Mishra and Aithal, 2021; Poudel, 2023).

There is also a striking contrast between alternative bricks and conventional bricks in terms of the energy consumed during the production process and the associated carbon and PM2.5

emissions (Table 9). For example, less than one-third of energy (coal) is needed to produce alternative bricks compared to clay-fired bricks. This less energy consumption translates into avoided emissions of 77% to 80% of CO₂ and over 90% of PM_{2.5} emissions when alternative bricks are used for building construction. Further, there is also a much lower social cost of PM_{2.5} associated with alternative bricks than with clay-fired brick production. It is estimated that the social cost of alternative bricks is NRs 0.29 for CSEB, and NRs 0.26 for HCB compared to NRs 6.81 for clay-fired bricks from ZZK. These energy and environmental benefits associated with alternative bricks are important in promoting them in the country.

Table 9: Avoided coal, CO₂, and PM_{2.5} emissions, and social cost of alternative building materials wrt clay-fired brick from selected kiln technologies

	Avoided energy (%)	Avoided emissions (%)		Social cost of PM _{2.5} (NRs)
	Coal	CO ₂	PM _{2.5}	
CSEB	69	77	92.6	0.29
HCB	72	80	93.4	0.26

Sources: (Fordham, 2021; Maithel et al., 2017) and authors' calculations.

5.4 International trade implications of coal replacement with domestic biomass in the brick industry

One of the key advantages of replacing imported coal with domestically supplied biomass fuels would be savings on import bills. Figure 8 shows the significant amount of coal imports that can be saved through switching from coal to domestic biomass pellets. The savings are calculated with an increasing percentage of coal substitution by domestic biomass pellets from the conventional kilns (FC BTK and ZZK) in 2021. It is estimated that the annual coal import savings range from NRs 2.1 billion to NRs. 6.1 billion if 10% of the coal is substituted by biomass pellets (Figure 10). Since all coal is imported in Nepal, these figures translate to about 6.5% to 18.9% of total coal import cost savings in 2021 (NTIP, 2023). It is important to note that the higher the percentage of coal substituted by biomass pellets, the higher the fuel cost savings. For example, if all coal is replaced by domestic biomass pellets for brick making from FC BTK and ZZK combined, close to two-thirds (NRs 20.6 billion) of total coal import costs can be reduced. Although all FC FTK and ZZK may not be able to run 100% on biomass pellets in the country for many practical reasons,

we did find one of the brick kilns in our rapid survey of kilns that relied 100% on biomass pellets in the western part of the country.

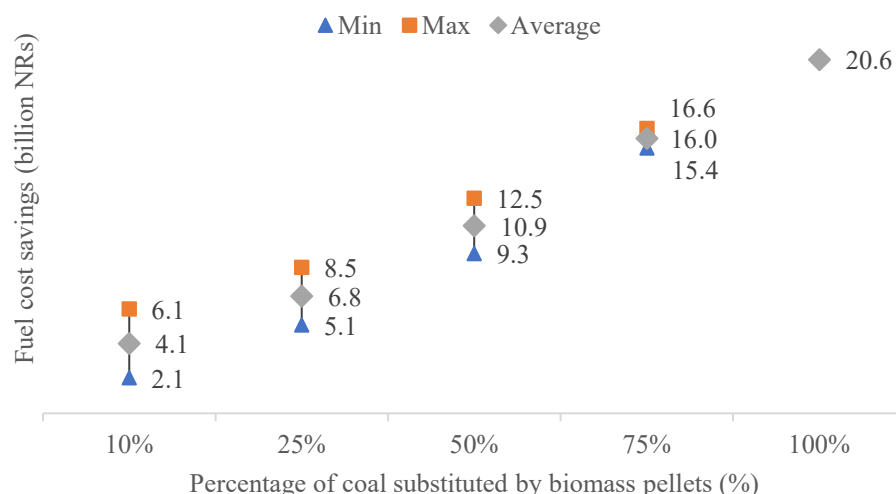


Figure 10. Annual fuel cost savings by switching from coal to biomass pellets with FC BTK and ZZK combined in 2022

Notes: The estimated total number of brick production in 2021 is 4.1 billion from FC BTK and ZZK. The ranges of values are dependent on the quality of coal (gross calorific value). Lower quality of coal yields lower cost savings. The calorific values considered are 4750 kcal/kg (low), 5,000 kcal/kg (average), and 5,500 kcal/kg (high) (SMSEE, 2017). The average price of coal for Nepal is NRs 43.4 per kg and the biomass pellets is NRs 12 per kg in 2022 from the survey.

6. Policy Discussion

6.1 Increased coal prices through a carbon tax

The economic analysis shows that energy-efficient technologies such as the VSBKs are the most economical fired clay brick production technology in terms of total private costs currently followed by ZZK and FC BTK. However, VSBK technology has low penetration in the country due to its high upfront costs and lack of skilled manpower to operate it. The ZZK technology, which is slightly more expensive (by NRs. 2 per brick) than VSBK is the most common technology in the country. Existing government policies caused a large-scale switching of inefficient fixed chimneys

to ZZK technologies over the last decade (Timilsina and Malla, 2023). If the price of coal is increased by 50% from the current level, the price of bricks produced from ZZK and FC BTK kilns, the most common technologies currently used in Nepal, would increase by 22%. This is equivalent to introducing a carbon tax of US\$100/ton of CO₂ (equivalent to NRs. 33,367 per ton of coal assuming the carbon content of coal is 70%). An increased coal price would help reduce CO₂ emissions and local environmental pollutants from brick kilns in two ways. First, it incentivizes brick producers to increase the fuel efficiency of coal-based technologies through the adoption of efficient kiln technologies like ZZK. Second, it incentivizes the replacement of coal with alternative fuels, such as biomass, particularly, pellets and briquettes. If a fixed chimney is replaced with Zig Zag technology, it will increase fuel efficiency by 9%, and thus reduce emissions of CO₂ by about 11% if coal is used. Brick industry operators suggest that the adoption of technology alone does not achieve efficiency, operational practices are more critical to achieve efficiency. For example, fuel-feeding practices are more responsible for improving efficiency than technology alone (Timilsina and Malla, 2023). Our analysis suggests that the cost of brick production from a Zig Zag chimney is slightly cheaper than that of a fixed chimney, and if the coal prices were to increase further, a Zig Zag chimney would be an even more attractive option. Coal prices can be increased through a new tax, such as a carbon tax or coal specific tax, or an environmental surcharge on coal. The increased price of coal will transmit to brick price. If the prices of bricks are higher in Nepal than in the neighboring countries, it would promote direct importation of bricks unless the government also imposes duties on imported bricks, although fired clay bricks are not imported at present. However, an increase in duties will invite reciprocity and trade confrontation.

Moreover, the improvement of energy efficiency has some barriers. The energy-efficient technologies are relatively capital-intensive and most brick factory owners in Nepal are middle-income entrepreneurs and might not be able to manage finance for installing energy-efficient technologies. It should also be noted that coal prices have almost doubled over the last three years. The factory-delivered prices of coal (i.e., transportation costs, import duties, and all taxes) varied from NRs. 20,000 to 30,000 per ton before the COVID-19 pandemic. By the end of 2022, it

increased up to NRs. 40,000 to 50,000 per ton almost due to the Russian invasion of Ukraine, the deteriorating exchange rate, and the rationing of coal exports by India.²⁰

The concern of price increase due to a carbon tax or adoption of cleaner technologies can be addressed by blending the carbon pricing policy with efficiency improvement (i.e., adoption of expensive cleaner technologies). Under this policy, increased fuel costs caused by emission pricing would be partially offset with capital subsidy for improving operational efficiency. Part of the revenue generated from the emission pricing is given back to the brick industry to reduce capital costs. Further analyses are needed to understand the economics of this policy blending.

6.2 Substitution of coal with biomass fuels

Sustainable biomass such as wood chips, pellets, briquettes, sawdust, and rice husks are normally considered alternative fuels for the brick industry. (Timilsina and Malla, 2023) find in their rapid survey that the use of biomass in brick chimneys is still commonplace in the Far Western region of Nepal. However, biomass can be used only in fixed chimneys and biomass fuels (briquettes, bagasse, and pellets) are mostly imported from India. Although biomass fuels have a lower heat value as compared to coal and therefore need more amount (i.e., tons) of biomass fuels than coal for the same amount of heat required by a brick kiln, the costs of bricks produced from biomass-fired kilns are lower than that of coal-fired kilns. This is because biomass is much cheaper than coal even if it is imported from India. If 50% coal is replaced by biomass pellets in a Zig Zag chimney, the production costs of bricks would drop by 11% to 18% depending upon the province under consideration. On average, the reduction would be 13% at the national level.

A question arises – why is it that the brick industries do not substitute coal with biomass if doing so is economically attractive? There are several reasons behind this. Supply constraint is the main reason. Although Nepal’s “Forest Act 2022” allows the extraction of forest products including fuelwood from the community as well as natural forests, the extraction from the natural forests is limited due to complex regulatory processes, lack of access to collection

²⁰ In the past ten years, non-coking coal export from India has decreased by half, from 2.4 million tons in 2013 to 1.2 million tons in 2022 (MOC, 2023).

machines/vehicles to natural forests and government ceiling on the amount of forest products (e.g., wood for producing woodchips) that can be extracted annually. While the extraction of forest products is simpler as compared to natural forests, the annual ceiling on the amount of forest products is also applicable for the community forests. These limitations imply that the supply of woodchips would be much smaller compared to the demand for the large-scale switching of coal to fuelwood in the brick industries. The limited supply and regulatory hurdles cause the fuel wood supply unreliable thereby discouraging brick kilns from switching to fuelwood. Large-scale switching of coal to wood chips looks unlikely in the brick industries as long as existing forest regulation is not changed. It needs strong political will and enhanced monitoring capacity.

Switching from coal to biomass for brick production can reduce PM_{2.5} emissions, but the extent of the reduction depends on the type of biomass used and the efficiency of the combustion process (kiln technology type). If biomass is efficiently burned, it produces fewer PM_{2.5} emissions compared to coal (US EPA, 1995). Cleaner and more homogenous biomass fuels, such as pellets, also tend to burn more completely than fuelwood. There are some technical constraints to switching biomass from coal in brick kilns. According to (Timilsina and Malla, 2023), some respondents to their rapid survey said that biomass fuels do not provide enough heat (measured in temperature) needed for quality brick production (i.e. technically advanced kilns including Zig Zag kilns). Fuels with gross calorific values (GCV) above 7,000-kilo calories (kcal) per kg are needed to attain a temperature of more than 900 C for good quality bricks, whereas biomass fuels can provide less than 5,000 kcal/kg.²¹ On the other hand, any type of biomass fuel can be used in fixed kilns. In western Nepal, many brick producers use biomass fuels in their fixed kilns.

6.3 Electric kilns could be the ideal technology, but it is not available in Nepal

Electric kilns would be an ideal option for brick production in Nepal due to the clean hydropower supply in Nepal. Electric kilns not only eliminate environmental pollution from the brick industry but also help improve trade balance and foreign currency reserves by substituting the imported coal used in the brick industry with domestically produced hydroelectricity. However, such a

²¹ <https://www.ecostan.com/calorific-value>

technology is not available commercially. There are some experimental or pilot projects either operational or planned to use electric kilns for brick-firing, such as the Kortemark plant in Belgium (Wienerberger, 2023), Uttendorf plant in Austria (NEFI, 2023), a full understanding of technical and economic viability of an electric kiln might take years. (Geres et al., 2021) project that economically viable GHG-neutral brick production would be available in developed economies like Germany by 2050. Another pathway for carbon-neutral brick production is the use of green hydrogen. However, the technical feasibility and economic viability of this technology are yet to be tested.

Considering the availability of clean electricity and the series of environmental and fiscal problems caused by imported coal, Nepal should explore the opportunities to use electric kilns for brick production. Since the technology is not yet commonplace, pilot programs on such technology should be undertaken. During the rapid survey (Timilsina and Malla, 2023), brick producers showed a strong interest in electric kilns. Some producers are willing to host pilots if they can get financial and technical support from the government or development partners.

6.4 Alternate (not-fired) bricks

Policies to promote alternative bricks would be a solution to reduce CO₂ emissions and other local pollutants from the brick industry in Nepal. At present, there are two alternative bricks in Nepal: unfired or uncooked bricks and cement blocks/bricks. The former is produced with a mix of clay, sand, stone dust, and some chemicals (for binding), whereas the latter is produced from cement. Since these bricks are not as strong as traditional bricks, they face marketing challenges. Many producers started to produce alternative bricks, mostly medium and small entrepreneurs, however, they face market barriers. Consumers are skeptical about the durability of such bricks. They argue that traditional bricks can survive thousands of years whereas alternative bricks may not survive for the lifetime of their buildings. While small producers are continuing production of alternative bricks, they are also uncertain about their future if the market does not develop (Timilsina and Malla, 2023). To develop the market for alternative bricks, the necessary policies needed are i) keeping the coal price at least from falling below the current level by collecting the difference as environmental levy if coal prices drop, ii) imposing an import duty to avoid the direct import of

conventional bricks, iii) providing investment subsidies, tax breaks or arrange carbon credits for alternative bricks, and iv) developing markets for alternative bricks through mandates and regulation and awareness, e.g., governments must use alternative bricks in public buildings and infrastructures, temples, and other public facilities, and local municipalities can also mandate residential buildings to use alternative bricks.

7. Concluding Remarks

The brick industry in Nepal is one of the primary sources of CO₂ emissions and local air pollutants. The main fuel used in the brick industry is coal, which is imported, mainly from neighboring India. Coal is responsible for about one-third of the current national CO₂ emissions from fossil fuel sources. The reduction of CO₂ emissions from the brick industry is crucial for meeting the country's nationally determined contribution by 2030 and net-zero emission target by 2045. Similarly, brick kilns are one of the main sources of local air pollution, particularly PM_{2.5}, around major cities where concentrations of PM_{2.5} exceed WHO standards by many folds. This study offers an economic analysis, from both private and social perspectives, of various alternatives to reduce coal consumption and corresponding emission reductions from the brick industry. The main data for the economic analysis is collected from primary sources through a field survey. The paper also offers policy discussion to reduce emissions from the brick industry in Nepal.

The economic analysis finds the costs of brick production vary across the provinces due to variations in input variables (e.g., cost of fuels, labor, land). The production costs of bricks produced from fixed chimney varies between NRs 16 and NRs 20 per brick across the provinces. In the case of the zig-zag chimney, which is more fuel efficient than the fixed chimney, the production costs are between NRs 17 and NRs 19 per brick. If a carbon tax of US\$100/tCO₂ is introduced, it would increase the current coal price by about 50%, thereby increasing the national average costs of fixed chimney bricks and zig-zag chimney bricks by 24% and 26%, respectively. If the use of coal in brick kilns is replaced with biomass fuels, the costs of bricks would reduce significantly apart from the reduction of CO₂ emissions. If 50% of coal in a zig-zag chimney is replaced with biomass pellets, the national-level average costs of bricks would decrease by 13%. Moreover, the replacement of imported coal with domestic biomass would cause significant

savings on the country's import bills and foreign reserves. For example, if all coal is replaced by domestic biomass pellets for brick making from fixed and zig-zag kilns currently operating in Nepal, about NRs 20.6 billion of import bills could be saved annually.

Local air pollutants, particularly PM_{2.5} from the brick industry are a major health challenge in urban areas in Nepal as it is estimated to cause 600 deaths annually and incur US\$ 46 million in costs on public health across the country (Eil et al., 2020). This study estimates that the production of brick from fixed and zig-zag kilns would cost society, through PM_{2.5} emission, about NRs 3 to NRs 9 <are these millions or per capita or per brick?> with a higher value in the highly populous Eastern region and a low value in the relatively low-populous far western region.

Several policy implications can be derived from the economic analysis. Pricing instruments (i.e., environmental fiscal policy), such as carbon tax could incentivize brick producers to reduce the use of coal. Government regulations encouraging brick kilns to switch from coal to biomass (wood chips, briquettes, pellets) would reduce emissions, particularly CO₂, as biomass obtained from sustainable forestry is carbon neutral. However, the limited and unreliable supply of forest biomass discourages a large-scale switching of coal to biomass in the brick industry. Unless private entrepreneurs are encouraged to enter into the biomass supply chain, the constraints on biomass fuel supply to the brick industry will not be resolved. Pricing policies may not work as long as the biomass supply constraints exist. This is evident from the fact that coal prices almost doubled over the last 3 years, switching to biomass fuels from coal has not happened. Hence, the environmental fiscal policy should consider a carbon tax on coal together with subsidies to cleaner technologies and domestic biomass fuel suppliers.

Since the brick industry has already faced prices as high as NRs 50/kg of coal (for the best quality), it might be possible to fix the price at this level for now. Even if international prices of coal drop, it can be sold to the brick factories at this rate. The difference between this price ceiling and the purchase price of coal might go to funds that can be used to subsidize alternative fuels and technologies. Moreover, governments or enterprises should try to tap international finance for GHG-reducing coal technologies through different channels, such as the Green Climate Fund. Technical Assistance, grants, and soft loans from development partners should be used to incentivize cleaner technologies.

Electric kilns would be an ideal solution for reducing CO₂ and other air pollutants from the brick industry in Nepal because the entire electricity in the country is hydro-based. Electric kilns

not only eliminate environmental pollution from the brick industry but also help improve the trade balance and foreign currency reserves by substituting the imported coal used in the brick industry with domestically produced hydroelectricity. However, such a technology is not available commercially. Piloting of electric kilns is critical to explore the potential of replacing coal with hydroelectricity. Brick factories are showing a huge interest in it. There is enormous interest from the government side as well because it enhances domestic hydro consumption for production activities thereby reducing GHG emissions (Timilsina and Malla, 2023). The pilot projects can be developed in partnerships between the governments, the private sector, and academia. Research and academia can publicize the findings from the pilots to benefit a wider audience.

An environmental-fiscal policy aiming to increase coal prices should also account for potential adverse economic implications, such as directly importing bricks. Brick factories we surveyed told us that bricks are cheaper in India than in Nepal because of increased coal prices. They argued if coal can be imported from the United States, why not import bricks from China or India? Nepal currently imports basic agricultural commodities (e.g., rice, onions, vegetables, fish) despite its potential to produce these commodities. If imported commodities are cheaper than domestically produced counterparts, it is not possible to prevent imports. Obviously, the importation of bricks causes losses in domestic employment and foreign currency reserves and deteriorates the trade balance.

Alternative bricks could be a solution, although not an ideal one. However, it faces market barriers due to their inferior quality as compared to regular bricks and perceptual biases among consumers. To develop the market for alternative bricks, several policies should be enacted together, including increasing (or keeping at the present level) coal prices through a carbon tax; imposing an import duty to avoid the direct import of bricks; providing investment subsidies, tax breaks, or arranging carbon credits for alternative bricks; and developing markets for alternative bricks through mandates and regulation and awareness.

References

Abbas, A., Sajid, M.B., Sajid, J., Ahmed, N., 2023. Forecasting environmental and social benefits of adopting cleaner technologies in Indian brick manufacturing industry. *Energy Sustain Dev.* 72, 202-211.

- Ahmad, F., Shah, G.M., Pradhan, B.B., Dixit, R., 2022. Towards an Environment-and-Worker Friendly Brick Kiln Sector in Punjab, Pakistan. Working Paper. International Centre for Integrated Mountain Development (ICIMOD), Kathmandu.
- Bhattarai, J., Ghale, D.B., Chapagain, Y.P., Bohara, N.B., Duwal, N., 2018. Study on the physical and mechanical properties of ancient clay brick samples of Kathmandu Valley, Nepal. *Tribhuvan University Journal* 32, 1-18.
- BiofuelCircle, 2022. Can Coal be Replaced with Biomass Briquettes? BiofuelCircle, Pune.
- Budhathoki, P., Duwal, N., Bhattarai, J., 2018. Investigation on physico-sintering and mechanical properties of clay tiles available in Kathmandu valley of Nepal. *Asian Journal of Chemistry* 30, 2515-2520.
- Budhwar, R., Bihari, V., Mathur, N., Srivastava, A., Kumar, S., 2003. DNA-protein crosslinks as a biomarker of exposure to solar radiation: a preliminary study in brick-kiln workers. *Biomarkers* 8, 162-166.
- Build Better Nepal, 2018. Competition - Interlocking Bricks (CSEB) vs Fired Brick - Building 1m2 Wall. Build Up Nepal, Kathmandu, <https://www.youtube.com/watch?v=wU2spFktjAA>.
- CBS, 2014. National Population and Housing Census 2011. Volume 04, NHHC2011. Central Bureau of Statistics, Kathmandu.
- CBS, 2022a. National Industrial Survey 2019/20: National Report. Central Bureau of Statistics (CBS), Government of Nepal, Kathmandu.
- CBS, 2022b. Statistical Yearbook 2021. Central Bureau of Statistics (CBS), Kathmandu.
- CEN, 2009. Baseline Study of Brick Kilns in Nepal. Clean Energy Nepal (CEN), Kathmandu.
- Chapagain, Y.P., Sapkota, S., Ghale, D.B., Bohara, N.B., Duwal, N., Bhattarai, J., 2020. A case study on mineralogy and physico-mechanical properties of commercial bricks produced in Nepal. *SN Appl. Sci.* 2, 14.
- Deore, M., Maudgal, M., Maithel, S., Kumar, S., 2019. Market Transformation towards Energy Efficiency in Brick Sector: A Strategic Blueprint, From Vision to Mission. Bureau of Energy Efficiency (BEE), Indo-German Energy Programme (GIZ), Greentech Knowledge Solutions Private Limited (GKSPL), New Delhi.
- DOI, 2022. Micro, Cottage and Small Industry Statistics (in Nepali). Department of Industry (DOI), Government of Nepal (GON), Kathmandu.
- DOMG, 2022. Annual Report of Department of Mines and Geology. Annual Report No. 13 DMG. Department of Mines and Geology (DOMG), Kathmandu.
- DUDBC, 2017. Design Catalogue for Reconstruction of Earthquake Resistance Houses: Volume II. Catalog for Alternative Construction Materials and Technologies. Department of Urban Development and Building Construction (DUDBC), Ministry of Urban Development, Government of Nepal, Kathmandu.

- DUDBC, 2023. Specifications of Building Construction (Civil) Works. DUDBC, Ministry of Urban Development, Government of Nepal, Kathmandu.
- EIA, 2022. Coal Explained: Types of Coal. Energy Information Administration (EIA), Washington, DC.
- Eil, A., Li, J., Baral, P., Saikawa, E., 2020. Dirty Stacks, High Stakes: An Overview of Brick Sector in South Asia. The World Bank, Washington, DC.
- EPIC INDIA, 2018. Using Emissions Markets to Manage the Social Costs of Fossil Fuels: A Handbook. Energy Policy Institute at the University of Chicago (EPIC India), New Delhi.
- FNCCI, 2017. Factsheet: Energy Efficiency for Brick Industry. Federation of Nepalese Chambers of Commerce and Industry (FNCCI), Kathmandu.
- Fordham, M., 2021. Build Up Nepal - Carbon Evaluation of CSEB Blocks. Carbon Evaluation of Compressed Stabilised Earth Blocks (CSEB) in Nepal. Max Fordham LLP, London.
- Geres, R., Lausen, J., Weigert, S., 2021. Roadmap for a greenhouse gas neutrality Brick industry in Germany: A path to climate neutrality for the industry by 2050 (in German, machine translated to English by google). FutureCamp Climate GmbH, Munich.
- Heierli, U., Maithel, S., 2008. Brick by Brick: The Herculean Task of Cleaning UP the Asian Brick Industry. Swiss Agency for Development and Cooperation (SDC), New Delhi.
- ICIMOD, 2019a. Brick Sector in Nepal: Fact Sheet. International Centre for Integrated Mountain Development (ICIMOD), Kathmandu.
- ICIMOD, 2019b. Burnt Clay Brick Sector in India. ICIMOD, Kathmandu.
- ICIMOD, 2019c. Gauging the Bankability of the Brick Sector in Nepal. ICIMOD, Kathmandu.
- IEA, 2022. Coal 2022: Analysis and Forecast to 2025. International Energy Agency (IEA), Paris.
- IEA, 2023a. Coal 2023: Analysis and forecast to 2026. IEA, Paris.
- IEA, 2023b. Driving Down Coal Mine Methan Emissions: A Regulatory Roadmap and Toolkit. IEA, Paris.
- IHME, 2020. Global Burden of Disease Collaborative Network. Global Burden of Disease Study 2019. Dataset. Institute for Health Metrics and Evaluation (IHME), Seattle, Washington.
- ILO, MOLESS, 2022. National Occupational Safety and Health Profile for Nepal 2022. International Labour Organization (ILO) and Ministry of Labour, Employment and Social Security (MOLESS) Kathmandu.
- ILO, UNCEF, CBS, 2020. Report on Employment Relationships in the Brick Industry in Nepal. ILO; United Nations Children's Fund (UNICEF); Central Bureau of Statistics (CBS), Kathmandu.

- Islam, M.R., Li, T., Mahata, K., Khanal, N., Werden, B., Giordano, M.R., Praveen Puppala, S., Dhital, N.B., Gurung, A., Saikawa, E., Panday, A.K., Yokelson, R.J., DeCarlo, P.F., Stone, E.A., 2022. Wintertime Air Quality across the Kathmandu Valley, Nepal: Concentration, Composition, and Sources of Fine and Coarse Particulate Matter. *ACS Earth and Space Chemistry* 6, 2955-2971.
- Jayarathne, T., Stockwell, C.E., Bhawe, P.V., Praveen, P.S., Rathnayake, C.M., Islam, M.R., Panday, A.K., Adhikari, S., Maharjan, R., Goetz, J.D., DeCarlo, P.F., Saikawa, E., Yokelson, R.J., Stone, E.A., 2018. Nepal Ambient Monitoring and Source Testing Experiment (NAMaSTE): emissions of particulate matter from wood- and dung-fueled cooking fires, garbage and crop residue burning, brick kilns, and other sources. *Atmos. Chem. Phys.* 18, 2259-2286.
- Kamyotra, J.S., 2015. Brick Kilns in India. Presentation. Central Pollution Control Board (CPCB), New Delhi.
- Kaushik, R., Khaliq, F., Subramanayaan, M., Ahmed, R.S., 2012. Pulmonary dysfunctions, oxidative stress and DNA damage in brick kiln workers. *Human & Experimental Toxicology* 31, 1083-1091.
- Khisroon, M., Khan, A., Imran, M., Zaidi, F., Ahmadullah, Fatima, S.H., 2018. Biomonitoring of DNA damage in individuals exposed to brick kiln pollution from Peshawar, Khyber Pakhtunkhwa, Pakistan. *Arch. Environ. Occup. Health* 73, 115-120.
- Kumar, S., Maithel, S., 2016. Introduction to Brick Kilns and Specific Energy Consumption Protocol for Brick Kilns Greentech Knowledge Solutions Pvt. Ltd., New Delhi.
- Lakho, N., Zardari, M., 2016. Structural Properties of Baked Clay Bricks Fired with Alternate Fuels. *Engineering* 8, 676-683.
- Maithel, S., 2013. Evaluating Energy Conservation Potential of Brick Production in India, A Report Prepared for the SAARC Energy Centre, Islamabad. Greentech Knowledge Solutions Pvt Ltd, New Delhi.
- Maithel, S., Kumar, S., Lalchandani, D., 2014. Factsheets about Brick Kilns in South and South-East Asia. Greentech Knowledge Solutions Pvt Ltd (GKSPL), New Delhi.
- Maithel, S., Lalchandani, D., Kumar, S., Bhanware, P., Ahuja, S., Uma, R., Athalye, V., Ragavan, S., Joshi, R., Bond, T., Weyant, C., Baum, E., 2013. Zigzag Kiln Performance Assessment – 2012 & 2013. A Shakti Sustainable Energy Foundation Supported Initiative. Greentech Knowledge Solutions, New Delhi, Enzen Global Solutions, Bangalore, University of Illinois at Urbana-Champaign, Illinois, Clean Air Task Force, Boston, New Delhi.
- Maithel, S., Lalchandani, D., Malhotra, G., Bhanware, P., Uma, R., Ragavan, S., Athalye, V., Bindiya, K.R., Reddy, S., Bond, T., Weyant, C., Baum, E., Thoa, V.T.K., Phuong, N.T., Thanh, T.K., 2012. Brick Kilns Performance Assessment: A Roadmap for Cleaner Brick Production in India. Greentech Knowledge Solutions Pvt. Ltd., New Delhi, Enzen Global Solutions Pvt.

- Ltd., Bangalore, University of Illinois at Urbana-Champaign, Illinois, Clean Air Task Force, Boston, Entec AG, Hanoi, New Delhi.
- Maithel, S., Ravi, A., Kumar, S., 2017. Roadmap for Promoting Resource Efficient Bricks in India: A 2032 strategy. Greentech Knowledge Solutions Pvt. Ltd., New Delhi.
- Manandhar, U.M., Dangol, S.B., 2013. Study on evaluating energy conservation potential of brick production in SAARC countries. Kathmandu, Nepal: Min Energy Initiatives, Nepal and SAARC Energy Center, Islamabad.
- MinErgy, 2020. Concrete Hollow Block Production. MinErgy Nepal, Kathmandu.
- MinErgy, FNBI, 2015. Design Manual: Improved Fixed Chimney Brick Kiln. A technical guide on how to design Induced Draught and Natural Draught Zig-Zag Kilns MinErgy Pvt. Ltd., Federation of Nepalese Brick Industries (FNBI), Kathmandu.
- Mishra, A.K., Aithal, P., 2021. Socio-Economic Suitability of AAC Block in Nepal. Solid State Technology 64, 575-595.
- MOC, 2023. Export of coal and coke from India during last ten years. Ministry of Coal (MOC), Government of India, New Delhi.
- MOFE, 2018. Collection of Environment Standards 2075. Ministry of Forest and Environment (MOFE), Government of Nepal (GON), Kathmandu.
- MOFE, 2021. Nepal's Third National Communication to the United Nations Framework Convention on Climate Change (UNFCCC). MOFE, GON, Kathmandu.
- NEFI, 2023. Sustainable Milestone: Europe's Greenest Brick Production is Being Established in Uttendorf. New Energy for Industry (NEFI), Wien, Austria.
- Nepal, S., Mahapatra, P.S., Adhikari, S., Shrestha, S., Sharma, P., Shrestha, K.L., Pradhan, B.B., Puppala, S.P., 2019. A Comparative Study of Stack Emissions from Straight-Line and Zigzag Brick Kilns in Nepal. Atmosphere 10, 19.
- NSO, 2023. National Population and Housing Census 2021. National Report. National Statistics Office (NSO), Kathmandu.
- NTIP, 2023. Import Report By Commodity Countries [Dataset]. Nepal Trade Information Portal (NTIP), Ministry of Industry, Commerce and Supplies (MOICS), Kathmandu.
- Poudel, K., 2023. BEEN: Promoting Hollow Brick, New Spotlight Fortnightly. Keshab Poudel, Kathmandu.
- Prajapati, S., Malla, A., Adhikari, N.P., Shrestha, S.L., 2019. Technical and Economic Assessment for the Promotion of Non Carbonized Pellets in MSME Sector in Nepal. Journal of the Institute of Engineering 15, 261-266.
- Prasain, K., 2021. Coal shortage hits domestic brick and cement factories, The Kathmandu Post, Kathmandu.

- Premchander, S., Bloesch, U., Tuladhar, B., Raghunandan, D., 2011. External Review of Clean Building Technologies for Nepal: VSBK-CESEF Project 2008-2011. Government of Nepal (GON) and Swiss Agency for Development and Cooperation (SDC), Kathmandu.
- PSCST, 2015. Utilization of Paddy Straw as Fuel in Brick Kilns. Punjab State Council for Science & Technology (PSCST), Chandigarh.
- Rajaratnam, U., Athalye, V., Ragavan, S., Maithel, S., Lalchandani, D., Kumar, S., Baum, E., Weyant, C., Bond, T., 2014. Assessment of air pollutant emissions from brick kilns. *Atmospheric Environment* 98, 549-553.
- Rawal, R., Maithel, S., Shukla, Y., Rana, S., Gowri, G., Patel, J., Kumar, S., 2020. Thermal Performance of Walling Material and Wall Technology, Part-1. Shakti Sustainable Energy Foundation, Greentech Knowledge Solutions Pvt. Ltd., Centre for Advanced Research in Building Science and Energy, New Delhi and Ahmedabad.
- Raza, A., Ali, Z., 2021. Impact of air pollution generated by brick kilns on the pulmonary health of workers. *Journal of Health Pollution* 11, 210906.
- Raza, A., Qamer, M.F., Afsheen, S., Adnan, M., Naeem, S., Atiq, M., 2014. Particulate Matter Associated Lung Function Decline in Brick Kiln Workers of Jalalpur Jattan, Pakistan. *Pak. J. Zool.* 46, 237-243.
- Sadavarte, P., Rupakheti, M., Bhawe, P., Shakya, K., Lawrence, M., 2019. Nepal emission inventory – Part I: Technologies and combustion sources (NEEMI-Tech) for 2001–2016. *Atmos. Chem. Phys.* 19, 12953-12973.
- Sah, D.P., Chaudhary, S., Shakya, R., Sah, P.K., Mishra, A.K., 2019. Status of Brick Kilns Stack Emission in Kathmandu Valley of Nepal. *Journal of Advanced Research in Civil and Environmental Engineering* 6, 1-9.
- Sanjel, S., Khanal, S.N., Thygeson, S.M., Carter, W.S., Johnston, J.D., Joshi, S.K., 2017. Respiratory symptoms and illnesses related to the concentration of airborne particulate matter among brick kiln workers in Kathmandu valley, Nepal. *Ann. Occup. Environ. Med.* 29, 12.
- Seay, B., Adetona, A., Sadoff, N., Sarofim, M.C., Kolian, M., 2021. Impact of South Asian brick kiln emission mitigation strategies on select pollutants and near-term Arctic temperature responses. *Environmental Research Communications* 3, 061004.
- Shrestha, S., 2019. A case study of brick properties manufacture in Bhaktapur. *Journal of Science and Engineering* 7, 27-33.
- Shriraam, V., Srihari, R., Gayathri, T., Murali, L., 2020. Active case finding for Tuberculosis among migrant brick kiln workers in South India. *Indian Journal of Tuberculosis* 67, 38-42.
- Siraz, M.M.M., Rakib, M.D.A., Alam, M.S., Al Mahmud, J., Rashid, M.B., Khandaker, M.U., Islam, M.S., Yeasmin, S., 2023. Assessment of radionuclides from coal-fired brick kilns on the outskirts of Dhaka city and the consequent hazards on human health and the environment. *Nuclear Engineering and Technology*.

- SMSEE, 2017. Report on Brick Kiln Stack Emission Monitoring in Kathmandu Valley. SMS Environment and Engineering Pvt. Ltd. (SMSEE), Kathmandu.
- Stockwell, C.E., Christian, T.J., Goetz, J.D., Jayarathne, T., Bhawe, P.V., Praveen, P.S., Adhikari, S., Maharjan, R., DeCarlo, P.F., Stone, E.A., Saikawa, E., Blake, D.R., Simpson, I.J., Yokelson, R.J., Panday, A.K., 2016. Nepal Ambient Monitoring and Source Testing Experiment (NAMaSTE): emissions of trace gases and light-absorbing carbon from wood and dung cooking fires, garbage and crop residue burning, brick kilns, and other sources. *Atmos. Chem. Phys.* 16, 11043-11081.
- Subedi, S.K., 2020. Study of Characteristics of Bricks Produced in Kathmandu, Nepal. *American Journal of Civil Engineering* 8, 64-76.
- Tandon, S., Gupta, S., Singh, S., Kumar, A., 2017. Respiratory Abnormalities among Occupationally Exposed, Non-Smoking Brick Kiln Workers from Punjab, India. *Int. J. Occup. Environ. Med.* 8, 166-173.
- TERI, 2017. Resource Efficient Bricks (REB): Benefits of Producing Clay-fired REBs. The Energy and Resources Institute (TERI), New Delhi.
- Thygerson, S.M., Sanjel, S., Johnson, S., 2016. Occupational and environmental health hazards in the brick manufacturing industry in Kathmandu valley, Nepal. *Occupational Medicine & Health Affairs* 4, 248.
- Tibrewal, K., Venkataraman, C., Phuleria, H., Joshi, V., Maithel, S., Damle, A., Gupta, A., Lokhande, P., Rabha, S., Saikia, B.K., Roy, S., Habib, G., Rathi, S., Goel, A., Ahlawat, S., Mandal, T.K., Azharuddin Hashmi, M., Qureshi, A., Dhandapani, A., Iqbal, J., Devaliya, S., Raman, R.S., Lian, Y., Pandithurai, G., Kuppili, S.K., Shiva Nagendra, M., Mukherjee, S., Chatterjee, A., Najar, T.A., Jehangir, A., Singh, J., Sinha, B., 2023. Reconciliation of energy use disparities in brick production in India. *Nature Sustainability* 6, 1248-1257.
- Timilsina, G., Malla, S., 2023. Brick Production in Nepal: A Rapid Survey. Unpublished Document. World Bank, Washington, DC.
- US EPA, 1995. Compilation of Air Pollutant Emission Factors, Fifth Edition. US Environmental Protection Agency, Washington, DC.
- Valdes, H., Vilches, J., Felmer, G., Hurtado, M., Figueroa, J., 2020. Artisan Brick Kilns: State-of-the-Art and Future Trends. *Sustainability* 12, 7724.
- Vasić, M.V., Goel, G., Vasić, M., Radojević, Z., 2021. Recycling of waste coal dust for the energy-efficient fabrication of bricks: A laboratory to industrial-scale study. *Environ. Technol. Innov.* 21, 101350.
- WECS, 2022. Energy Sector Synopsis Report 2021/2022. Report No. WECS 03/077/078. Water and Energy Commission Secretariat (WECS), Government of Nepal (GON), Kathmandu.
- WECS, 2023. Energy Sector Synopsis Report 2023. WECS, GON, Kathmandu.

- Weyant, C., Athalye, V., Ragavan, S., Rajarathnam, U., Lalchandani, D., Maithel, S., Baum, E., Bond, T.C., 2014. Emissions from South Asian Brick Production. *Environmental Science & Technology* 48, 6477-6483.
- Weyant, C., Kumar, S., Maithel, S., Thompson, R., Floess, E., Ravi, A., Adhikari, S., Gautam, S., Shrestha, S., Baum, E., Bond, T., 2016. Emissions and energy performance in South Asian brick kilns: Measurement report. University of Illinois Urbana-Champaign, Civil & Environmental Engineering & Greentech Knowledge Solutions Pvt. Ltd. & Mountain Air Engineering & International Centre for Integrated Mountain Development (ICIMOD).
- WHO, 2021. WHO Global Air Quality Guidelines. Particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide. World Health Organization (WHO), Geneva.
- Wienerberger, 2023. 2022 Annual and Sustainability Report. Wienerberger AG, Vienna.
- World Bank, 2019. Nepal Environment Sector Diagnostic: Path to Sustainable Growth Under Federalism. Country Environmental Analysis. World Bank, Washington, DC.
- World Bank, 2021. The Global Health cost of PM2.5 Air Pollution. A Case for Action Beyond 2021. World Bank, Washington, DC.

Annexes

Annex I: Key Survey Dataset

Table A1.1: Prices of coal and biomass fuels, and brick weight by province*

	Koshi	Madhesh	Bagmati	Gandaki	Lumbini	Sudurpachim	Nepal (Weighted avg.)
Coal (NRs/kg)	41.7	40.0	46.0	45.5	41.8	45.5	43.4
Biomass (NRs/kg)							
Traditional	8.0	8.0	8.0	8.0	8.2	-	8.04
Briquettes/Pellets	-	-	-	-	-	12.0	
Brick weight (kg/brick)	3.917	3.375	2.500	2.750	2.613	2.529	2.874

Note: * Because there are only two brick kilns in Karnali province (ILO et al., 2020), we excluded this province from our study. Source: (Timilsina and Malla, 2023)

Table A1.2: Type of technology and fuel used by the surveyed kilns

	Technology	Fuel type
1	CK	Coal
2	ZZK ID	Coal and sawdust
3	ZZK ID	Coal
4	FC BTK	Coal, sawdust and rice husk
5	ZZK ND	Coal
6	ZZK ND	Coal
7	HK	Coal, sawdust and bagasse
8	ZZK ID	Coal, sawdust, firewood and rice husk
9	ZZK ND	Coal and sawdust
10	ZZK ND	Coal, sawdust and firewood
11	ZZK ID	Coal, sawdust, firewood and rice husk
12	ZZK ND	Coal
13	ZZK ND	Coal
14	ZZK ND	Coal and firewood
15	ZZK ID	Coal
16	FC BTK	Coal
17	FC BTK	Coal and pellet
18	ZZK ID	Coal
19	FC BTK	Pellet
20	ZZK ID	n.a.
21	ZZK ID	Coal
22	FC BTK	Coal

Note: CK is clamp kiln, FC BTK is fixed chimney Bull's trench kiln, ZZK is zig zag kiln, ID and ND are induced draft and natural draft, HK is Hoffman kiln and n.a. is not available.

Source: (Timilsina and Malla, 2023)

Annex II: Benchmark Dataset (parameters) Description and Sources

A2.1 Activity and Structure-Related Data Compilation

The Volume of brick production [$N_{r,g}(t)$]

Table A2.1: Volume of brick production ($N_{r,g}$) in 2018 and the number of operational brick kilns in 2019 by province

Province	Brick Production by brick quality ^a (million)					Number of kilns by employee size ^b			
	Grade A	Grade B	Grade C	Others	Total	Small	Medium	Large	Total
Koshi	140	63	39	20	262	7	36	42	85
Madesh	452	149	106	18	725	16	151	125	292
Bagmati	499	361	63	2	925	10	35	160	205
Gandaki	90	34	30	-	154	26	20	16	62
Lumbini	455	208	93	15	771	35	81	110	226
Karnali ^c	-	-	4	-	4	-	2	-	2
Sudurpachim ^c	95	61	37	5	198	16	39	39	94
Total	1731	876	372	60	3039	110	364	492	966

Source: (ILO et al., 2020)

Notes:

^a Grade A is well baked, shiny, standard size, uniformly yellow or red, of regular texture, and uniformly shaped. Such bricks are free from pebbles, gravel, or organic materials. Grade B bricks are slightly over-cooked and may have irregular shapes and textures in comparison to Grade A bricks. Grade C bricks are crumbling, small pieces, overlooked, and irregular in shape. Others include tiles and other materials.

^b Small kilns have employees of less than 50, medium kilns have employees between 50 and 100, and large kilns have employees of more than 100.

^c Brick production data for Karnali is not available. We assume the type of technology of two operational kilns in Karnali is CK and they produce Grade C bricks. Note that out of 41 million Grade C bricks produced in Karnali and Sudurpachim provinces combined, we assume 4.2 million [i.e., $(202/96) \times 2$] Grade C bricks are produced in Karnali, and the remaining 36.8 million Grade C bricks are produced in Sudurpachim. The value, 202, is the total number of bricks produced (Karnali + Sudurpachim), the value, 96, is the total number of kilns (Karnali + Sudurpachim), and the value, 2, is the total number of kilns in Karnali.

The percentage weight by brick quality type [$\bar{w}_{j,g}$]

Table A2.2: Estimated percentage weight by brick quality type ($\bar{w}_{j,g}$) and the number of brick kilns by technology type

Technology type (j)	Percentage weight by brick quality type (g) ^a				Brick kiln ^b	
	Grade A	Grade B	Grade C	Others	Number (N_j)	Share (%)
CK		74.1	100.0		125	9.3
FC BTK	75.0	23.8			899	66.6
ZZ FC BTK	24.1				289	21.4
VSBK		2.1			25	1.9
HK	0.7			72.7	8	0.6
TK	0.3			27.3	3	0.2
Total	100.0	100.0	100.0	100.0	1,349	100.0

Notes:

^a We estimated the percentage weight as $w_j = (N_j / \sum_j^m N_j) \times 100$. Note that $\forall j$ does not belong to J . We assume Grade A bricks are produced only by FCBTK, ZZ FCBTK, HK, and TK; Grade B bricks are produced by CK, FCBTK, and VSBK; Grade C bricks are produced only by CK; and others are produced by HK and TK. For example, the percentage weight (i.e., share) of FCBTK technology that produces Grade A bricks is 75%. That is, $75\% = [899 / (899 + 289 + 8 + 3)] \times 100$. Note that only four technologies (FCBTK, ZZ FCBTK, HK, and TK) produce Grade A bricks.

^b Data for the year 2018 taken from (ICIMOD, 2019a). Note that 7 MC BTK listed in (ICIMOD, 2019a) are not accounted for in our study. We assume all MC BTK is not in operation due to the government ban. We also assume that these weights (share) are the same for all provinces (r).

The average mass of clay-fired bricks [$M_r(t)$]

Table A2.3: Physical and mechanical characteristics of clay-fired bricks* in Nepal

Data source	Location	Size (cm ³)	D _B (g/cm ³)	W _{AC} (%)	S _C (MPa)	M (kg)
(Bhattarai et al., 2018) ^a	Kathmandu Valley		1.20 1.80	–	10.0 – 18.0 23.00	–
(Shrestha, 2019)	Kathmandu Valley	1,255	1.55 2.82	–	8.80 23.93	– 2.20 ^b
(Subedi, 2020)	Kathmandu Valley	1,315	1.34 1.90	–	4.00 12.00	– 2.03 ^c
(Budhathoki et al., 2018) ^d	Kathmandu Valley		1.53 2.15	–	4.20 27.26	–
(Chapagain et al., 2020)	Nepal		1.10 2.15	–	5 – 30 10.53	–
(Eil et al., 2020)	Nepal					2.00 ^e
(Thygeson et al., 2016)	Kathmandu Valley					1.50 – 2.00
(MinErgy and FNBI, 2015)	Kathmandu Valley					2.10
(CEN, 2009)	Nepal					2.03 (2.76) ^f
Gorkha Chinese Itta (2023)	Nepal	1,573			8.50	2.50 (2.00) ^g
Timilsina and Malla (2023)	Nepal					2.40 – 4.00 ^h

Notes:

* Based on the NS 1 2035 IS code guidelines, the machine-made standard clay-fired brick size in Nepal is 240 mm x 112 mm x 57 mm (=1,532 cm³), and the chimney-made standard clay-fired brick size is 229 mm x 112 mm x 55 mm (1,411 cm³), with a tolerance of ± 3 mm in each direction and the compressive strength of 3.5 N/mm² for both bricks (DUDBC, 2023) (see pp 59). However, based on available studies, there is no uniformity of brick size in Nepal as the size of bricks is different in various places inside and outside of the Valley. In general, bricks in Tarai are bigger in size whereas those within the Kathmandu Valley and in hill regions are smaller in size.

^a Values reported are based on ancient bricks used in temples and historical monuments.

^b The size of the average sample brick is 1,255 cm³ and the density of the majority of sample bricks is between 1.5 to 2.0 g/cm³. Assuming an average size is 1,255 cm³ and a density value is 1.75 g/cm³, the estimated weight of each brick is 2.2 kg (=1.75*1255/1000).

^c The average weight of each clay-fired standard dry brick is 2.025 kg (~2.03 kg) for Bagmati province. This is the average value of 6 sample brick weights produced by FCBTK in Kathmandu Valley (i.e., 1.95; 2.18; 2.21; 1.71; 2.1 kg).

^d Values reported are based on tiles. Tiles are generally used as coverings for floors, walls, facades, or roofs and can range from simple square tiles to complex mosaics.

^e Based on first-class (high quality) brick with great variation in height of the bricks.

^f Brick weight is 2.03 kg in Kathmandu Valley and 2.76 kg outside of Kathmandu Valley.

^g Weight of each solid brick is 2.5 kg and the hollow brick is 2.0 kg taken from <http://gorkhachineseitta.com/>.

^h Based on a rapid survey of 22 brick kilns across the country. In general, the bricks in the Tarai region are about 1.25 to 1.5 times larger than the bricks in Kathmandu Valley.

A2.2 Intensity-Related Data Compilation [$I_j^{th}(t)$ and $I_j^{el}(t)$]

There is a wide variation in thermal specific energy consumption (SEC) within and across different brick kiln technologies. Even with the same kiln technology, many factors influence thermal SEC for firing clay bricks, including the combination of external and internal fuels used (e.g., coal, sawdust, rice husk, briquette, pellet, coal ash, carbonaceous industrial wastes etc.), the quality of the fuels used (e.g., fuels with different calorific values), the type and weight of desired clay-fired bricks (e.g., solid, perforated or hollow bricks) and the kiln's overall heat loss and the air leakage management. For example, in Pakistani brick kilns, additives such as agricultural residue, sawdust, *gutka*, poultry waste, and rubber, are used with coal that ranges from low 2.5% to as high as 100% (Ahmad et al., 2022).

One of the main advantages of a ZZK is its lower SEC as compared to that of FCBTK. On average, in comparison to conventional FCBTK, the ZZKs in India require about 20% less energy (Maithel et al., 2013). Likewise, the VSBK consumes about 35% less fuel as compared to FCBTK because of better combustion and heat recovery, and low heat losses (Maithel et al., 2014).

Table A2.4: Compilation of Specific Energy Consumption (SEC) of different brick kilns in South Asia (MJ/kg of clay-fired brick)

Source ^a	CK	DDK	FCBTK	ZZK ND	ZZK ID	VSBK	HK	TK	Quality ^b
1 ^c	1.58	-	1.30	1.19	-	-	-	-	P
2 ^d	2.10	2.97	1.30	1.06	1.03	0.80	1.36	1.40	S [3]
3	2.90	-	1.30	1.30	1.10	0.80	1.20	1.20	S (V)
4	2.00-4.00	2.00-4.00	1.00-1.50	1.0-1.2	-	0.80-1.50	2.00-4.00	1.5-2.00	S (V)
5 ^{d,e}	-	-	0.99-1.19	-	0.60-1.06	-	-	-	P
6 ^f	2.00	-	1.30	1.13	1.18	-	-	1.60	S [16]
7 ^g	-	-	1.12-1.21	-	-	-	1.95-2.12	-	P
8 ^{d,h}	-	-	3.21	1.90	-	-	-	-	P
9 ⁱ	2.91	-	1.12	-	-	0.84	-	-	P
10 ^{d,j}	1.33	1.90	1.17	1.07	1.14	-	-	-	P
11 ^{d,k}	-	-	-	1.10	-	-	-	-	P
12 ^l	1.38-1.92	2.80-3.14	0.95-1.82	0.91-1.15	-	0.90	1.21-1.52	-	U
13	2.10	2.97	1.30	1.06	1.03	0.80	1.20-1.36	1.40	S [16]
14 ^d	2.36	-	1.25	1.16	0.92	0.83	1.50	-	S (V)
15	1.50-2.00	2.50-3.00	1.00-1.40	1.0-1.2	-	0.75-1.00	-	-	S (U)
16 ^m	2.91	-	1.22	1.06	-	0.95	-	1.46	P
17 ^d	-	-	1.16	0.92	-	0.83	-	-	S

18	2.00-4.50	-	1.10-1.50	-	-	0.7-1.0	-	1.10-2.50	p
19									

Notes:

^a Data sources are 1 (Tibrewal et al., 2023); 2 (Abbas et al., 2023); 3 (Eil et al., 2020); 4 (Valdes et al., 2020); 5 (Nepal et al., 2019); 6 (Deore et al., 2019); 7 (TERI, 2017); 8 (FNCCI, 2017); 9 (Kumar and Maithel, 2016); 10 (Weyant et al., 2016); 11 (MinErgy and FNBI, 2015); 12 (Kamyotra, 2015); 13 (Maithel et al., 2014); 14 (Manandhar and Dangol, 2013); 15 (Maithel, 2013); 16. (Maithel et al., 2012); 17 (Premchander et al., 2011); 18 (Heierli and Maithel, 2008); and 19 Rapid Survey study.

^b P is the primary source; S is the secondary source; V is the various sources; and U is the source unknown. The number in the parenthesis is a reference number from a data source.

^c Fuel types and their share by kiln technology and by data source are as follows: For BTK: 0.19 MJ/kg (100% coal), 1.26 MJ/kg (coal dominated), 1.33 MJ/kg (biomass dominated), and 1.43 MJ/kg (100% biomass); For ZZK: 1.02 MJ/kg (100% coal), 1.24 MJ/kg (coal dominated), and MJ/kg 1.32 (100% biomass); CK: 1.64 MJ/kg (low capacity), 1.51 MJ/kg (medium capacity), and 1.07 MJ/kg (high capacity). The SEC for average internal fuels (boiler ash and agriculture waste) is 0.48 MJ/kg.

^d Nepal-specific data sources.

^e For BTK: 1.17 MJ/kg (90% coal and 10% rice husk), 1.04 MJ/kg (70% coal and sawdust, and 30% rice husk), 1.19 MJ/kg (100% bagasse briquette), and 0.99 MJ/kg (40% coal and 60% bagasse briquette.; For ZZK: 0.60 MJ/kg (100% coal), 0.91 MJ/kg (100% coal), and 1.06 MJ/kg (43% coal, 32% coal and rice husk and 25% rice husk as internal fuel).

^f For ID ZZK, the value is for 25% perforation brick; and for TK, the value is for 60% perforation hollow brick.

^g All values are 100% coal-based. Electric SEC is excluded. BTK with perforated brick production is 1.06-1.16 MJ/kg.

^h Based on the combination of coal, sawdust, and bagasse. Values reported are in MJ/brick.

ⁱ Values for CK (100% fuelwood); BTK (100% coal), VSBK (100% coal as external fuel; boiler ash/wheat straw as internal fuels)

^j Based on a combination of coal and fuelwood.

^k (MinErgy and FNBI, 2015). 100% coal.

^l All is 100% coal-based. Also reported is 100% biomass-based for BTK as 1.33-1.95 MJ/kg of fired brick.

^m For BTK: 1.46 MJ/kg (coal, wood and tire), 1.12 MJ/kg (100% coal), and 1.1 MJ/kg (coal and coal slurry); For ZZK: 1.21 MJ/kg (coal and sawdust), and 1.03 MJ/kg (100% coal); For VSBK it is coal and coal slurry; for CK it is fuelwood; and for TK, it is coal ash and coal slurry. For TK, an additional 0.26 MJ/kg of brick (total 1.72 MJ/kg) of electric energy is used.

A2.3 Calorific Value of Fuels-Related Data Compilation

Coal is mainly used for firing commercial clay bricks in Nepal. Many classifications are used around the world with the main parameter being the coal rank. The ranks of coal, based on the carbon content (in decreasing order) are anthracite (86% - 97% carbon), bituminous (45% - 86% carbon), subbituminous (35% - 45% carbon), lignite (25% - 35% carbon) and peat (< 25% carbon) (IEA, 2022; EIA, 2022). The first three coal types are also known as hard coal. Coal is also classified by its intended use, e.g., thermal (or steam) and metallurgical applications. Thermal coal refers to hard coal used for purposes other than metallurgy. Metallurgical coal refers to coking coal, semi-soft coal, and pulverized injection coal, and they are high-quality coal mainly used to produce coke or to reduce coke consumption used in blast furnaces to make pig iron. Note that depending on its intended use, anthracite can be classified either as thermal coal (used for heat and electricity generation) or metallurgical coal (used for making coke). More than 80% of all coal produced today is steam coal, which is mainly used for heat and electricity generation (IEA, 2023b). Nearly 5% is lignite, which is also generally used for power generation. Coking coal makes up around 15% of global production and is mainly used in steelmaking.

The calorific value and impurity content are the main parameters defining the quality of thermal (steam) coal, whereas caking properties, resistance, and impurity content are the distinguishing characteristics of coking coal (IEA, 2022). The quality (calorific value) of thermal coal mainly depends on its moisture content, ash content, volatile matter, and fixed carbon content. In general,

lower moisture and ash contents and volatile matter, and higher fixed carbon correspond to higher calorific value of the coal. So, lignite has lower calorific value than sub-bituminous and bituminous coal, and (sub) bituminous coal has lower calorific value than anthracite. Normally, only the organic parts of the coal are burned during the combustion process, whereas carbon dioxide and water vapor are produced as by-products. The inorganic parts of coal usually do not burn but stay in the ash as by-products.

Table A2.5: Gross calorific values (kcal/kg) of commonly used fuels for brick making process in India and Nepal

Fuel type	(Weyant et al., 2016)	(Kumar and Maithel, 2016)	(PSCST, 2015)	(SMSEE, 2017)	(Prajapati et al., 2019)
Coal	6,483 (6,254) ^a	4,000 – 7,000	-	4,000 - 5,500 ^a	-
Wood chips	-	3,500 – 4,500	-	-	-
Sawdust	4,000 ^a	3,500 – 4,500	-	-	2,961 ^c
Coffee husk	-	4,300 – 4,500	-	-	-
Mustard stalk	4,007	-	3,985	-	-
Biomass briquettes ^b	-	-	3,155 ^c	-	3,277 ^c
Paper plant ash ^d	704	-	-	-	-
Iron plant ash ^d	2,789	-	-	-	-
Bagasse	3,166	-	-	-	-
Coal ash ^d	2,906	-	-	-	-
Paddy straw ^d	-	-	3,386	3,403 - 3,471 ^a	-
Cow dung ^d	-	-	2,831	-	-

Notes:

^a Nepal-specific value. ^b Biomass pellets and briquettes are environmentally friendly alternative fuels to coal in brick making. Other advantages include higher calorific value when pelletized these biomass fuels and the creation of a market. Pellets and briquettes are produced by compacting loose biomass such as sawdust, crop waste, tree branches, rice husk, straw, and bagasse. The calorific value of high-quality briquettes ranges between 18 and 24 MJ/kg (BiofuelCircle, 2022). ^c Paddy and mustard ratio of 50:50.

^d Mainly used as internal fuels during clay preparation and molding. ^e Reported values are for Nepal and expressed in MJ/kg. We use the conversion factor of 1MJ = 239 kcal. Sawdust is a mixture of 70% sawdust and 30% wood chips. Biomass briquettes (pellets) are sawdust pellets imported from India. The average cost of sawdust is NRs 10 per kg and the pellets made from sawdust and imported from India are NRs 26 per kg.

Most of the coal consumed in Nepal is imported from India, but some is also imported from as far as Indonesia, the United States, South Africa, and Australia. A small quantity of low-quality lignite is produced domestically in the Lumbini province (Dang and Palpa districts) of Nepal. In the past 12 years (2010-2021), about 98,000 tons of lignite have been excavated (DOMG, 2022). This is equivalent to an average of 8,128 tons per year and it is less than 1% of total coal imported each year (WECS, 2022). Most of these imported coals are relatively low-rank thermal (steam) coal mainly used for firing clay bricks. The gross calorific of imported coal varies widely from a low value of 4,000 kcal/kg (Indian coal) to a high value of 5,500 kcal/kg (Indonesian coal) (SMSEE, 2017).²² The gross calorific values of commonly used fuels for firing clay bricks in India and

²² A local coal supplier and distributor (AKTraders) located in Kapilvastu district reported gross calorific value of 7,300 kcal/kg for American coal, 1,330 - 6,200 kcal/kg for Indian coal, and 4,200 kcal/kg for Indonesia. This information cited from <https://nepalcoal.com/> and extracted on October 20, 2023.

Nepal, such as coal, biomass fuels, and industrial wastes are presented in Table A2.5. Some brick kilns in Nepal use coal mixed with other biomass fuels such as sawdust, wood chips, and rick husk, especially during the initial firing in the brick-making process. In India, a few coal-based brick kilns also use LPG as an alternative to biomass fuels, such as fuelwood, for initial firing because it results in good combustion and decreases fuel consumption which in turn reduces energy costs (ICIMOD, 2019b). For example, when 50% of fuelwood is substituted with LPG for initial firing, fuel cost can be reduced by 60%. Furthermore, compared with fuelwood, LPG significantly produces fewer air pollutant emissions and provides instantaneous heat that improves production quality.

A2.4 Emission Factors and Air Pollutant Emissions-Related Data Compilation

The brick industry in Nepal is one of the main sources of GHG emissions and local air pollutants. This industry is responsible for about two-thirds of GHG emissions (CO₂eq) from the manufacturing and construction sector (2,262 Gg), and 10% of the country's total GHG emissions from the energy sector (14,703 Gg) in 2011 (MOFE, 2021).

Many factors influence the quantity of air pollutant emissions from the brick industry, such as brick kiln technology type, fuel use, and kiln operating conditions. Comparing the air pollutant emissions across different fuels and different operating conditions requires normalization, either to a unit of fuel consumed or to a unit of energy consumed, or a comparison based on brick production. In our study, we use normalized emission factors based on brick production. To estimate emissions, an emission factor is multiplied by the corresponding activity data, the production output of bricks in weight, and the energy contained in a mass of fuel combusted. Note that the conversion units can significantly increase the uncertainty of the results when the reported values vary widely.

The common air pollutants from brick kilns and their impact on human health and climate are summarized in Table A2.6.

Table A2.6: Air pollutant emissions and its impact on human health and climate associated with brick kilns

	Description	Impact
PM	Particulate matter (PM) refers to inhalable particles, composed of sulfate, nitrates, ammonia, sodium chloride, black carbon, mineral dust, or water. The most common PMs are PM _{2.5} and PM ₁₀ relevant for health.	Health: Mortality associated with cardiovascular (e.g., heart disease), cerebrovascular (e.g., stroke), respiratory (e.g., COPD) and lung cancer.
CO	Carbon monoxide (CO) is a colorless, odorless, and tasteless toxic gas produced by the incomplete combustion of carbonaceous fuels such as coal in brick kilns.	Health: Hospital admissions and emergency room visits associated with difficulties in breathing, exhaustion, dizziness, and other flu-like symptoms.
NO ₂	Nitrogen dioxide (NO ₂) is a reddish-brown gas that is soluble in water, and a strong oxidant. It results from high-temperature combustion of fuels in processes such as brick kilns.	Health: Hospital admissions and emergency room visits linked to asthma and other respiratory conditions. Climate: An important ozone precursor and acid rain

SO ₂	Sulfur dioxide (SO ₂) is a colorless gas with a sharp odor that is readily soluble in water. It is predominantly derived from the combustion of fossil fuels such as coal.	Health: Hospital admissions and emergency room visits linked to asthma and other respiratory conditions. Climate: Acid rain
BC	Black carbon (BC) is a major component of PM _{2.5} and it is sometimes referred to as soot. Its main sources are from incomplete combustion of fossil fuels, biofuels, and biomass.	Health: Short- and long-term exposure to BC is associated with cardiovascular health effects and premature mortality.
CO ₂	Carbon dioxide (CO ₂) is the primary source of man-made GHG emitted mainly from the combustion of fossil fuels.	Climate: Global warming

Source: (WHO, 2021)

The compilation of normalized average emission factors for brick kilns in South Asia and Nepal is summarized in Tables A2.7 – A2.9.

Table A2.7: Normalized average production-based emission factors for brick kilns (g/kg of fired brick)

Kiln type	Data source	Particulate Matter				Flue gas pollutant			
		SPM	PM ₁₀	PM _{2.5}	BC	SO ₂	NO _x	CO	CO ₂
CK	2	1.60	1.60	0.97	0.290	3.60	0.07	-	-
	4	-	-	0.64	0.001	1.05	-	6.50	-
DDK	1	1.56	-	0.97	0.290	n.d.	-	5.78	282
	3	-	-	0.49	0.190	-	-	13.20	-
	6	-	-	-	0.192	<0.001	-	13.20	271
	8	1.56	-	-	-	<0.001	-	5.01	526
FCBTK	1	0.86	-	0.18	0.130	0.66	-	2.25	115
	2	2.20	0.36	0.21	0.170	3.60	0.07	-	-
	3	-	-	0.18	0.157	-	-	2.03	-
	4 ^a	-	-	0.30	0.100	2.80	-	1.20	-
	5	0.86	-	0.18	0.130	0.66	-	2.25	115
	6	-	-	-	0.151	0.65	-	2.23	124
	7	1.18	-	-	0.130	-	-	2.00	131
	8	0.89	-	-	-	-	-	3.63	179
	9	-	-	0.20	0.003	1.2	-	-	96
ZZK ND	1	0.26	-	0.13	0.040	0.32	-	1.47	103
	2	0.31	0.20	0.12	0.035	0.88	0.06	-	-
	3	-	-	0.13	0.020	-	-	0.60	-
	4 ^b	-	-	0.18	0.030	1.70	-	1.30	-
	5	0.26	-	0.13	0.040	0.32	-	1.47	103
	6	-	-	-	0.010	0.13	-	0.96	103
	7	0.22	-	-	0.010	-	-	0.29	105
	8	0.22	-	-	-	0.06	-	0.35	119
ZZK FD	3	-	-	0.05	0.011	-	-	1.23	-
	4 ^a	-	-	0.24	0.030	1.70	-	2.20	-
	7	0.24	-	-	0.020	-	-	1.62	98
	8	0.24	-	-	-	0.24	-	2.04	96
	9	-	-	0.10	0.10	0.90	-	-	82
VSBK	1	0.11	-	0.09	0.002	0.54	-	1.84	70
	2	0.31	0.18	0.11	0.002	0.54	-	-	-
	3	-	-	0.07	0.002	-	-	1.95	-
	5	0.11	-	0.09	0.002	0.54	-	1.84	70
	6	-	-	-	0.002	0.59	-	2.19	69
	7	0.15	-	-	0.001	-	-	1.84	71

	8	0.09	-	-	-	0.10	-	4.14	118
HK	2	0.31	0.24	0.15	0.003	0.62	0.03	-	-
	5	0.25	-	-	-	-	-	-	85
	7	0.29	-	-	-	-	-	-	100
TK	1	0.31	-	0.18	n.d.	0.72	-	2.45	166
	2	0.23	0.15	0.09	0.001	0.67	0.15	-	-
	3	-	-	0.24	0.001	-	-	4.50	-
	5	0.31	-	0.18	n.d.	0.66	-	2.45	166
	7	0.24	-	-	n.d.	-	-	3.31	166
	8	0.31	-	-	-	0.72	-	2.28	149

Table A2.8: Normalized average fuel-based emission factors for brick kilns (g/kg of fuel)

Kiln type	Data source	Particulate Matter			Flue gas pollutant				
		SPM	PM _{2.5}	BC	SO ₂	NO ₂	CO	CO ₂	CH ₄
CK	4	-	1.10 (2.3)	0.01	11.2 (18.3)	-	48.8 (70.3)	-	-
	11 ^a	-	-	0.0172	13.00	0.297	70.90	2102	19.50
	12 ^a	-	10.66	-	-	-	-	-	-
DDK	1	9.30	5.75	1.71	<0.001	-	34.36	1680	-
	3	-	3.00	1.10	-	-	78.60	-	-
BTK ND	1	14.15	3.03	2.44	10.45	-	41.14	2182	-
	4 ^a	-	6.8 (5.0)	3.2 (2.0)	66 (33.60)	-	28.1 (39.3)	-	-
	9	-	3.80	0.60	22.00	-	-	1633	-
ZZK ND	1	5.82	2.66	0.83	3.86	-	32.40	2017	-
	4	-	(4.70)	(0.70)	(44.70)	-	(34.10)	-	-
ZZK FD	4 ^a	-	5.70	0.60	39.3	-	51.10	-	-
	9	-	3.10	0.40	24.00	-	-	1981	-
	11 ^a	-	-	0.112	13.00	0.082	10.10	2620	0.0872
	12 ^a	-	15.11	-	-	-	-	-	-
VS BK	1	1.93	1.80	0.046	7.86	-	39.90	1375	-
TK	1	2.02	1.19	<0.001	4.73	-	16.19	1097	-
	3	-	1.60	0.01	-	-	29.60	-	-

Table A2.9: Normalized average energy-based emission factors for brick kilns (g/MJ of energy)

Kiln type	Data source	Particulate Matter			Flue gas pollutant					
		SPM	PM _{2.5}	BC	SO ₂	NO _x	CO	CO ₂	CH ₄	NMVOCs
CK	4 ^a	-	0.1	0.00025	0.95	-	4.0	-	-	-
	9	0.16	-	0.103	0.32	0.056	3.2	90	0.48	0.056
DDK	1	0.54	0.33	0.10	<0.001	-	1.99	97	-	-
	8	0.54	-	-	<0.1	-	5.17	181	-	-
	10	0.53	-	0.098	1.21	0.023	1.95	95.08	-	0.098
BTK ND	1	0.65	0.14	0.10	0.50	-	1.78	92	-	-
	4 ^a	-	0.3 (0.2)	0.10	2.4 (1.4)	-	1 (1.7)	-	-	-
	8	0.66	-	-	0.39	-	2.96	140	-	-
	10	0.91	-	0.10	0.39	0.059	1.54	100.8	0.0042	0.055
ZZK ND	1	0.25	0.12	0.04	0.30	-	1.39	91	-	-
	4	-	(0.2)	(0.02)	(1.6)	-	(1.2)	-	-	-
	8	0.21	-	-	0.06	-	0.32	113	-	-
	10	0.21	-	0.009	0.152	0.069	0.27	99.05	0.0049	0.064
ZZK FD	4 ^a	-	0.2	0.02	1.5	-	2.0	-	-	-
	8	0.23	-	-	0.23	-	1.96	92	-	-
	10	0.23	-	0.019	0.15	-	1.57	94.7	0.0049	0.064
VS BK	1	0.16	0.13	0.0023	0.95	-	2.28	99	-	-

	8	0.10	-	-	0.11	-	4.39	126	-	-
	10	0.19	-	0.00125	0.11	0.093	2.3	88.12	0.0066	0.086
HK	10	0.24	-	0.0025	0.48	0.024	1.5	83.3	-	-
TK	1	0.21	0.12	-	0.49	-	1.67	113	-	-
	8	0.21	-	-	0.49	-	1.56	109	-	-
	10	0.17	-	0.0007	0.48	0.107	2.36	118.8	-	-

Sources:

1. (Maithel et al., 2012); 2. (Eil et al., 2020); 3. (Weyant et al., 2014); 4. (Weyant et al., 2016); 5. (ICIMOD, 2019c); 6. (Seay et al., 2021); 7. (Maithel et al., 2014); 8. (Rajarithnam et al., 2014); 9. (Nepal et al., 2019); 10. (Abbas et al., 2023); 11. (Stockwell et al., 2016); 12. (Jayarathne et al., 2018).

Notes:

^a Nepal-specific value. ^b Indian kilns. The main fuel used by kiln technology for data sources is as follows. 1: BTK (mainly coal, but also rubber tires and wood logs); ZZK (coal, sawdust); VSBK (coal, coal powder, and slurry as internal fuels); DDK (biomass fuelwood); and TK (coal). 2: All kilns are coal-based. 3: BTK (coal, coal slurry, wood, rubber); ZZK (coal, sawdust, petcoke); VSBK (coal, coal slurry, coal powder); DDK (wood) and TK (coal ash and coal slurry). 4: CK (coal, bagasse, iron and paper plant ash, coal fines); Indian BTK (coal, mustard stalk, wood); Nepalese BTK (coal); Indian ZZK (coal, petcoke, sawdust); Nepalese ZZK (coal and sawdust). 5: Information not available. Note that values for ZZK, VSBK, HK, and TK are calculated based on the value of FCBTK. 6: same as in data source 3. 7: CK (coal and biomass); BTK (coal, biomass, petcoke, rubber tires); ZZK (coal, biomass, petcoke, rubber tires); VSBK (coal); HK (coal, biomass); TK (coal, petcoke). 8: CK (coal, biomass); BTK, ZZK, and VSBK are coal. 9: BTK (coal, rick husk, and briquette) and ID ZZK (coal, rick husk, sawdust). 10: Same as in 2 and 7. 11 and 12: CK (Coal and fuelwood) and ZZK (coal and bagasse).

For source 4, CK values are for large clamps and the figures in the parenthesis are for small clamps. For others, values are for Nepal and the values in the parenthesis are for India.